````
---
title: "Lithko - Team UD | Predicting the Next Construction Boom"
---

```{r}
# CHUNK 1 — Load libraries
library(tidyverse)
library(lubridate)
library(janitor)
library(readxl)
library(broom)
library(corrplot)
library(scales)
library(knitr)
library(kableExtra)
```

```{r}
# CHUNK 1b — Shared helper functions
# These are reused across all 5 sector analyses (Data Center, Warehouse,
# Apartment, Office, Manufacturing). Defined once here to avoid repetition.

calculate_error_metrics <- function(actual, predicted) {
  tibble(
    MAE  = mean(abs(actual - predicted), na.rm = TRUE),
    MAPE = mean(abs((actual - predicted) / actual) * 100, na.rm = TRUE),
    RMSE = sqrt(mean((actual - predicted)^2, na.rm = TRUE))
  )
}
```

```{r}
# Chunk 2 — Load Lithko internal data
jobs <- read_csv("job data", show_col_types = FALSE) %>% clean_names()
labor <- read_csv("labor hours data", show_col_types = FALSE) %>% clean_names()
customers <- read_csv("customer data", show_col_types = FALSE) %>% clean_names()
business_unit <- read_csv("business unit data", show_col_types = FALSE) %>% clean_names()
region <- read_csv("Region data", show_col_types = FALSE) %>% clean_names()
```

```{r}
# Chunk 3 — Standardize dates and build true construction start
jobs <- jobs %>%
  mutate(creation_date = as_datetime(creation_date))

labor <- labor %>%
  mutate(work_date = as_date(work_date))

true_job_start <- labor %>%
````
````

````
  group_by(job_id) %>%
  summarise(
    true_start_date = min(work_date, na.rm = TRUE),
    .groups = "drop"
  )

jobs
labor
true_job_start
```

```{r}
# Chunk 4 — Build enriched Lithko dataset
jobs_enriched <- jobs %>%
  left_join(true_job_start, by = "job_id") %>%
  left_join(
    business_unit %>%
      rename(
        bu_name = name,
        bu_city = city,
        bu_state = st_cd
      ),
    by = "ou_id"
  ) %>%
  left_join(
    region %>% select(ou_id, region),
    by = "ou_id"
  ) %>%
  left_join(customers, by = "cus_id")
```

```{r}
# Chunk 5 — Filter to warehouse and data center
## 5. Filter Target Segments and Assign Quarter

jobs_filtered <- jobs_enriched %>%
  filter(project_type %in% c("WAREHOUSE", "DATA CENTER")) %>%
  mutate(
    quarter = floor_date(true_start_date, "quarter")
  )
```

```{r}
# Chunk 6 — Build quarterly internal activity dataset
## 6. Build Quarterly Internal Activity Dataset

job_starts_quarterly <- jobs_filtered %>%
  group_by(project_type, quarter) %>%
  summarise(
````

    job_starts = n_distinct(job_id),
    .groups = "drop"
)

labor_hours_quarterly <- labor %>%
  left_join(
    jobs_filtered %>% select(job_id, project_type),
    by = "job_id"
) %>%
  mutate(
    quarter = floor_date(work_date, "quarter")
) %>%
  filter(!is.na(project_type)) %>%
  group_by(project_type, quarter) %>%
  summarise(
    labor_hours = sum(hrs, na.rm = TRUE),
    .groups = "drop"
)

baseline_quarterly <- job_starts_quarterly %>%
  left_join(labor_hours_quarterly, by = c("project_type", "quarter")) %>%
  arrange(project_type, quarter)
```

```{r}
# Chunk 7 — Remove incomplete final quarter
## 7. Remove Incomplete Final Quarter

max_work_date <- max(labor$work_date, na.rm = TRUE)
last_quarter_start <- floor_date(max_work_date, "quarter")
last_complete_quarter <- last_quarter_start %m-% months(3)

baseline_quarterly_complete <- baseline_quarterly %>%
  filter(quarter <= last_complete_quarter)
```

```markdown
## Chunk 8 - Data Center Segment Analysis

This section focuses on Lithko’s historical data-center-related construction activity and evaluates whether selected external indicators improve early-signal forecasting performance relative to an internal-only baseline. Following project feedback, the goal is not to build a black-box prediction system, but to test whether a small set of explainable external variables improves directional planning support.
```

```{r}
# CHUNK 9 — Load Data Center External Indicators

data_center_spending <- read_csv("monthly-spending-data-center-us.csv", show_col_types = FALSE) %>%

````
  clean_names()

tcu_raw <- read_excel("Capacity Utilization Total Index.xlsx", sheet = "Monthly") %>%
  clean_names()

pce_raw <- read_excel("Personal consumption expenditures (implicit price deflator).xlsx", sheet = "Quarterly")
%>%
  clean_names()

electricity_raw <- read_excel("Industrial Production Utilities Electric and Gas Utilities.xlsx", sheet = "Monthly")
%>%
  clean_names()
```

```{r}
# CHUNK 10 — Clean External Indicators

data_center_spending <- data_center_spending %>%
  rename(
    observation_date = day,
    dc_spending = monthly_spending_on_data_center_construction_in_the_united_states
  ) %>%
  mutate(
    observation_date = as_date(observation_date)
  ) %>%
  select(observation_date, dc_spending)

tcu_clean <- tcu_raw %>%
  mutate(
    observation_date = as_date(observation_date)
  ) %>%
  select(observation_date, tcu)

pce_clean <- pce_raw %>%
  rename(
    pce_deflator = dpcerd3q086sbea
  ) %>%
  mutate(
    observation_date = as_date(observation_date)
  ) %>%
  select(observation_date, pce_deflator)

electricity_clean <- electricity_raw %>%
  rename(
    utility_production = ipg2211a2n
  ) %>%
  mutate(
    observation_date = as_date(observation_date)
  ) %>%
````

  select(observation_date, utility_production)
```

```{r}
# CHUNK 11 — Convert External Indicators to Quarterly Frequency

min_q <- min(baseline_quarterly_complete$quarter, na.rm = TRUE)
max_q <- max(baseline_quarterly_complete$quarter, na.rm = TRUE)

to_quarterly_mean <- function(df, value_col, out_name) {
  df %>%
    mutate(quarter = floor_date(observation_date, "quarter")) %>%
    group_by(quarter) %>%
    summarise(value = mean(.data[[value_col]], na.rm = TRUE), .groups = "drop") %>%
    rename(!!out_name := value) %>%
    filter(quarter >= min_q, quarter <= max_q)
}

dc_spending_q <- to_quarterly_mean(data_center_spending, "dc_spending", "dc_spending")
tcu_q <- to_quarterly_mean(tcu_clean, "tcu", "tcu")
utility_q <- to_quarterly_mean(electricity_clean, "utility_production", "utility_production")

pce_q <- pce_clean %>%
  mutate(quarter = floor_date(observation_date, "quarter")) %>%
  group_by(quarter) %>%
  summarise(pce_deflator = mean(pce_deflator, na.rm = TRUE), .groups = "drop") %>%
  filter(quarter >= min_q, quarter <= max_q)
```

```{r}
# CHUNK 12 — Build Data Center Quarterly Internal Dataset

data_center_quarterly <- baseline_quarterly_complete %>%
  filter(project_type == "DATA CENTER") %>%
  select(quarter, labor_hours, job_starts) %>%
  arrange(quarter)

data_center_quarterly
```

```{r}
# CHUNK 13 — Merge Quarterly External Indicators

external_indicators_q <- dc_spending_q %>%
  full_join(tcu_q, by = "quarter") %>%
  full_join(utility_q, by = "quarter") %>%
  full_join(pce_q, by = "quarter") %>%
  arrange(quarter)

````
external_indicators_q
```

```{r}
# CHUNK 14 — Quick Visual Check of External Indicators

external_indicators_long <- external_indicators_q %>%
  pivot_longer(
    cols = -quarter,
    names_to = "indicator",
    values_to = "value"
  )

ggplot(external_indicators_long, aes(x = quarter, y = value, color = indicator)) +
  geom_line(linewidth = 1) +
  facet_wrap(~ indicator, scales = "free_y", ncol = 2) +
  labs(
    title = "Quarterly External Indicators",
    x = "Quarter",
    y = "Value",
    color = "Indicator"
  ) +
  theme_minimal(base_size = 12)
```

```markdown
## External Indicator Design

We selected external indicators to better reflect the economic and infrastructure conditions that may influence data-center-related construction activity. These indicators were converted to quarterly frequency to align with Lithko's internal reporting cadence. Because the goal is directional planning support rather than exact causal attribution, the indicators are evaluated as potential leading signals to test whether they improve forecast accuracy relative to an internal-only baseline.
```

```{r}
# CHUNK 15 — Build Data Center Modeling Table

dc_model_data <- data_center_quarterly %>%
  left_join(dc_spending_q, by = "quarter") %>%
  left_join(pce_q, by = "quarter") %>%
  left_join(tcu_q, by = "quarter") %>%
  left_join(utility_q, by = "quarter") %>%
  arrange(quarter)

glimpse(dc_model_data)
head(dc_model_data)
```
````
````

```{r}
# CHUNK 16 — Create Lagged Versions of Internal and External Variables

dc_model_data <- dc_model_data %>%
  arrange(quarter) %>%
  mutate(
    qtr = factor(
      paste0("Q", lubridate::quarter(quarter)),
      levels = c("Q1", "Q2", "Q3", "Q4")
    ),

    labor_hours_lag1 = lag(labor_hours, 1),
    labor_hours_lag2 = lag(labor_hours, 2),
    labor_hours_lag3 = lag(labor_hours, 3),
    labor_hours_lag4 = lag(labor_hours, 4),

    dc_spending_lag1 = lag(dc_spending, 1),
    dc_spending_lag2 = lag(dc_spending, 2),
    dc_spending_lag3 = lag(dc_spending, 3),
    dc_spending_lag4 = lag(dc_spending, 4),

    pce_deflator_lag1 = lag(pce_deflator, 1),
    pce_deflator_lag2 = lag(pce_deflator, 2),
    pce_deflator_lag3 = lag(pce_deflator, 3),
    pce_deflator_lag4 = lag(pce_deflator, 4),

    tcu_lag1 = lag(tcu, 1),
    tcu_lag2 = lag(tcu, 2),
    tcu_lag3 = lag(tcu, 3),
    tcu_lag4 = lag(tcu, 4),

    utility_production_lag1 = lag(utility_production, 1),
    utility_production_lag2 = lag(utility_production, 2),
    utility_production_lag3 = lag(utility_production, 3),
    utility_production_lag4 = lag(utility_production, 4)
  )

glimpse(dc_model_data)
```

```{r}
# CHUNK 17 — Create Horizon-Specific Forecasting Function
# Uses calculate_error_metrics() defined in CHUNK 1b.

fit_dc_horizon_models <- function(data, horizon, test_periods = 4) {

  horizon_data <- data %>%
    transmute(
      quarter,
```

```r
    qtr,
    labor_hours,
    labor_hours_lag = .data[[paste0("labor_hours_lag", horizon)]],
    dc_spending_lag = .data[[paste0("dc_spending_lag", horizon)]],
    pce_deflator_lag = .data[[paste0("pce_deflator_lag", horizon)]],
    tcu_lag = .data[[paste0("tcu_lag", horizon)]],
    utility_production_lag = .data[[paste0("utility_production_lag", horizon)]]
  ) %>%
  drop_na()

n_obs <- nrow(horizon_data)

if (n_obs < 8) {
  stop(paste("Not enough observations available for horizon", horizon))
}

test_n <- min(test_periods, max(2, floor(n_obs * 0.25)))
train_n <- n_obs - test_n

train_data <- horizon_data %>%
  slice(1:train_n)

test_data <- horizon_data %>%
  slice((train_n + 1):n_obs)

baseline_model <- lm(
  labor_hours ~ labor_hours_lag + qtr,
  data = train_data
)

augmented_model <- lm(
  labor_hours ~ labor_hours_lag +
    I(dc_spending_lag / 100000000) +
    pce_deflator_lag +
    tcu_lag +
    utility_production_lag +
    qtr,
  data = train_data
)

test_predictions <- test_data %>%
  mutate(
    baseline_forecast = predict(baseline_model, newdata = test_data),
    augmented_forecast = predict(augmented_model, newdata = test_data)
  )

baseline_metrics <- calculate_error_metrics(
  actual = test_predictions$labor_hours,
  predicted = test_predictions$baseline_forecast
```

```
  )

  augmented_metrics <- calculate_error_metrics(
    actual = test_predictions$labor_hours,
    predicted = test_predictions$augmented_forecast
  )

  list(
    horizon = horizon,
    horizon_data = horizon_data,
    train_data = train_data,
    test_data = test_data,
    baseline_model = baseline_model,
    augmented_model = augmented_model,
    predictions = test_predictions,
    baseline_metrics = baseline_metrics,
    augmented_metrics = augmented_metrics
  )
}
```

```{r}
# CHUNK 18 — Run True 1Q, 2Q, 3Q, and 4Q Ahead Models

dc_horizon_results <- list(
  `1Q Ahead` = fit_dc_horizon_models(dc_model_data, horizon = 1),
  `2Q Ahead` = fit_dc_horizon_models(dc_model_data, horizon = 2),
  `3Q Ahead` = fit_dc_horizon_models(dc_model_data, horizon = 3),
  `4Q Ahead` = fit_dc_horizon_models(dc_model_data, horizon = 4)
)

names(dc_horizon_results)
```

```{r}
# CHUNK 19 — Forecast Error Summary Table

dc_forecast_error_summary <- bind_rows(
  lapply(names(dc_horizon_results), function(h_name) {
    result <- dc_horizon_results[[h_name]]

    tibble(
      segment = "Data Center",
      forecast_horizon = h_name,
      baseline_MAE = result$baseline_metrics$MAE,
      enhanced_MAE = result$augmented_metrics$MAE,
      baseline_MAPE = result$baseline_metrics$MAPE,
      enhanced_MAPE = result$augmented_metrics$MAPE,
      baseline_RMSE = result$baseline_metrics$RMSE,
```

````
      enhanced_RMSE = result$augmented_metrics$RMSE
    )
  })
) %>%
  mutate(
    MAE_lift_pct = ((baseline_MAE - enhanced_MAE) / baseline_MAE) * 100,
    MAPE_lift_pct = ((baseline_MAPE - enhanced_MAPE) / baseline_MAPE) * 100,
    RMSE_lift_pct = ((baseline_RMSE - enhanced_RMSE) / baseline_RMSE) * 100
  )

dc_forecast_error_summary %>%
  mutate(
    across(
      c(
        baseline_MAE, enhanced_MAE,
        baseline_RMSE, enhanced_RMSE
      ),
      ~ comma(round(.))
    ),
    across(
      c(
        baseline_MAPE, enhanced_MAPE,
        MAE_lift_pct, MAPE_lift_pct, RMSE_lift_pct
      ),
      ~ percent(. / 100, accuracy = 0.1)
    )
  ) %>%
  kable(
    caption = "Data Center Forecast Error Summary by Horizon"
  ) %>%
  kable_styling(full_width = FALSE)
```

```{r}
# CHUNK 20 — Lift in Accuracy Summary Table

dc_lift_summary <- dc_forecast_error_summary %>%
  select(
    forecast_horizon,
    baseline_RMSE,
    enhanced_RMSE,
    RMSE_lift_pct,
    baseline_MAE,
    enhanced_MAE,
    MAE_lift_pct
  )

dc_lift_summary %>%
  mutate(
````

````
    baseline_RMSE = comma(round(baseline_RMSE)),
    enhanced_RMSE = comma(round(enhanced_RMSE)),
    RMSE_lift_pct = percent(RMSE_lift_pct / 100, accuracy = 0.1),
    baseline_MAE = comma(round(baseline_MAE)),
    enhanced_MAE = comma(round(enhanced_MAE)),
    MAE_lift_pct = percent(MAE_lift_pct / 100, accuracy = 0.1)
  ) %>%
  kable(
    caption = "Data Center Lift in Accuracy from External Indicators"
  ) %>%
  kable_styling(full_width = FALSE)
```

```{r}
# CHUNK 21 — One Coefficient Table Per Regression

# 1Q Ahead
dc_1q_result <- dc_horizon_results[["1Q Ahead"]]

tidy(dc_1q_result$baseline_model) %>%
  mutate(
    estimate = round(estimate, 4),
    std.error = round(std.error, 4),
    statistic = round(statistic, 3),
    p.value = round(p.value, 4)
  ) %>%
  kable(caption = "Data Center 1Q Ahead — Baseline Coefficients") %>%
  kable_styling(full_width = FALSE)

tidy(dc_1q_result$augmented_model) %>%
  mutate(
    estimate = round(estimate, 4),
    std.error = round(std.error, 4),
    statistic = round(statistic, 3),
    p.value = round(p.value, 4)
  ) %>%
  kable(caption = "Data Center 1Q Ahead — Augmented Coefficients") %>%
  kable_styling(full_width = FALSE)


# 2Q Ahead
dc_2q_result <- dc_horizon_results[["2Q Ahead"]]

tidy(dc_2q_result$baseline_model) %>%
  mutate(
    estimate = round(estimate, 4),
    std.error = round(std.error, 4),
    statistic = round(statistic, 3),
    p.value = round(p.value, 4)
````

```
  ) %>%
  kable(caption = "Data Center 2Q Ahead — Baseline Coefficients") %>%
  kable_styling(full_width = FALSE)

tidy(dc_2q_result$augmented_model) %>%
  mutate(
    estimate = round(estimate, 4),
    std.error = round(std.error, 4),
    statistic = round(statistic, 3),
    p.value = round(p.value, 4)
  ) %>%
  kable(caption = "Data Center 2Q Ahead — Augmented Coefficients") %>%
  kable_styling(full_width = FALSE)


# 3Q Ahead
dc_3q_result <- dc_horizon_results[["3Q Ahead"]]

tidy(dc_3q_result$baseline_model) %>%
  mutate(
    estimate = round(estimate, 4),
    std.error = round(std.error, 4),
    statistic = round(statistic, 3),
    p.value = round(p.value, 4)
  ) %>%
  kable(caption = "Data Center 3Q Ahead — Baseline Coefficients") %>%
  kable_styling(full_width = FALSE)

tidy(dc_3q_result$augmented_model) %>%
  mutate(
    estimate = round(estimate, 4),
    std.error = round(std.error, 4),
    statistic = round(statistic, 3),
    p.value = round(p.value, 4)
  ) %>%
  kable(caption = "Data Center 3Q Ahead — Augmented Coefficients") %>%
  kable_styling(full_width = FALSE)


# 4Q Ahead
dc_4q_result <- dc_horizon_results[["4Q Ahead"]]

tidy(dc_4q_result$baseline_model) %>%
  mutate(
    estimate = round(estimate, 4),
    std.error = round(std.error, 4),
    statistic = round(statistic, 3),
    p.value = round(p.value, 4)
  ) %>%
```

```
  kable(caption = "Data Center 4Q Ahead — Baseline Coefficients") %>%
  kable_styling(full_width = FALSE)

tidy(dc_4q_result$augmented_model) %>%
  mutate(
    estimate = round(estimate, 4),
    std.error = round(std.error, 4),
    statistic = round(statistic, 3),
    p.value = round(p.value, 4)
  ) %>%
  kable(caption = "Data Center 4Q Ahead — Augmented Coefficients") %>%
  kable_styling(full_width = FALSE)

dc_coefficients_by_horizon <- bind_rows(
  tidy(dc_1q_result$baseline_model) %>%
    mutate(model_type = "Baseline", forecast_horizon = "1Q Ahead"),
  tidy(dc_1q_result$augmented_model) %>%
    mutate(model_type = "Augmented", forecast_horizon = "1Q Ahead"),

  tidy(dc_2q_result$baseline_model) %>%
    mutate(model_type = "Baseline", forecast_horizon = "2Q Ahead"),
  tidy(dc_2q_result$augmented_model) %>%
    mutate(model_type = "Augmented", forecast_horizon = "2Q Ahead"),

  tidy(dc_3q_result$baseline_model) %>%
    mutate(model_type = "Baseline", forecast_horizon = "3Q Ahead"),
  tidy(dc_3q_result$augmented_model) %>%
    mutate(model_type = "Augmented", forecast_horizon = "3Q Ahead"),

  tidy(dc_4q_result$baseline_model) %>%
    mutate(model_type = "Baseline", forecast_horizon = "4Q Ahead"),
  tidy(dc_4q_result$augmented_model) %>%
    mutate(model_type = "Augmented", forecast_horizon = "4Q Ahead")
) %>%
  mutate(
    significance = case_when(
      p.value < 0.01 ~ "***",
      p.value < 0.05 ~ "**",
      p.value < 0.10 ~ "*",
      TRUE ~ ""
    )
  )
```

```{r}
# CHUNK 22 — Indicators Tested Summary by Horizon

dc_indicator_tested_summary <- dc_coefficients_by_horizon %>%
  filter(
```

~~~
    model_type == "Augmented",
    !term %in% c("(Intercept)", "labor_hours_lag"),
    !str_detect(term, "^qtr")
  ) %>%
  mutate(
    indicator = case_when(
      term == "I(dc_spending_lag/1e+08)" ~ "dc_spending_lag",
      TRUE ~ term
    ),
    direction = case_when(
      estimate > 0 ~ "Positive",
      estimate < 0 ~ "Negative",
      TRUE ~ "Neutral"
    ),
    included_in_final_model = if_else(p.value < 0.10, "Yes", "No")
  ) %>%
  select(
    forecast_horizon,
    indicator,
    estimate,
    p.value,
    significance,
    direction,
    included_in_final_model
  )

dc_indicator_tested_summary %>%
  mutate(
    estimate = round(estimate, 4),
    p.value = round(p.value, 4)
  ) %>%
  kable(
    caption = "Data Center Indicators Tested by Horizon"
  ) %>%
  kable_styling(full_width = FALSE)
```

```{r}
# CHUNK 23 — RMSE Comparison Plot by Horizon

dc_rmse_plot_data <- dc_forecast_error_summary %>%
  select(
    forecast_horizon,
    baseline_RMSE,
    enhanced_RMSE
  ) %>%
  pivot_longer(
    cols = c(baseline_RMSE, enhanced_RMSE),
    names_to = "model_type",
~~~

```
    values_to = "RMSE"
  ) %>%
  mutate(
    model_type = recode(
      model_type,
      baseline_RMSE = "Baseline",
      enhanced_RMSE = "Enhanced"
    ),
    forecast_horizon = factor(
      forecast_horizon,
      levels = c("1Q Ahead", "2Q Ahead", "3Q Ahead", "4Q Ahead")
    )
  )

ggplot(dc_rmse_plot_data, aes(x = forecast_horizon, y = RMSE, fill = model_type)) +
  geom_col(position = "dodge") +
  scale_y_continuous(labels = comma) +
  labs(
    title = "Data Center RMSE by Forecast Horizon",
    subtitle = "Baseline vs Enhanced Models",
    x = "Forecast Horizon",
    y = "RMSE",
    fill = "Model"
  ) +
  theme_minimal(base_size = 12)
```

```{r}
# CHUNK 24 — Actual vs Predicted Plot for 3Q Ahead Test Period

dc_3q_plot_data <- dc_horizon_results[["3Q Ahead"]]$predictions %>%
  select(quarter, labor_hours, baseline_forecast, augmented_forecast) %>%
  pivot_longer(
    cols = c(labor_hours, baseline_forecast, augmented_forecast),
    names_to = "series",
    values_to = "value"
  ) %>%
  mutate(
    series = recode(
      series,
      labor_hours = "Actual",
      baseline_forecast = "Baseline Forecast",
      augmented_forecast = "Enhanced Forecast"
    )
  )

ggplot(dc_3q_plot_data, aes(x = quarter, y = value, color = series, linetype = series)) +
  geom_line(linewidth = 1.2) +
  scale_y_continuous(labels = comma) +
```

````
  labs(
    title = "Data Center 3Q Ahead Forecast: Actual vs Predicted",
    subtitle = "Test Period Comparison",
    x = "Quarter",
    y = "Labor Hours",
    color = "",
    linetype = ""
  ) +
  theme_minimal(base_size = 12)
```

```{r}
# CHUNK 25 — Load Warehouse External Indicators

ecom_raw <- read_csv("E-Commerce Retail Sales.csv", show_col_types = FALSE) %>%
  clean_names()

pnfi_raw <- read_excel("Private_Nonresidential_Fixed_Investment.xlsx", sheet = "Quarterly") %>%
  clean_names()

tlnrescons_raw <- read_excel("Total Construction Spending-Nonresidential in the United States .xlsx", sheet =
"Monthly") %>%
  clean_names()

indpro_raw <- read_excel("INDPRO(manufacturing).xlsx", sheet = "Monthly") %>%
  clean_names()

dgs10_raw <- read_excel("Interest_Rate_10Y.xlsx", sheet = "Daily") %>%
  clean_names()
```

```{r}
# CHUNK 26 — Clean Warehouse External Indicators

ecom_clean <- ecom_raw %>%
  rename(
    observation_date = observation_date,
    ecom_sales = e_commerce_retail_sales
  ) %>%
  mutate(
    observation_date = mdy(observation_date)
  ) %>%
  select(observation_date, ecom_sales)

pnfi_clean <- pnfi_raw %>%
  rename(
    observation_date = observation_date,
    pnfi = private_nonresidential_fixed_investment
  ) %>%
````

```
  mutate(
    observation_date = ymd(observation_date)
  ) %>%
  select(observation_date, pnfi)

tlnrescons_clean <- tlnrescons_raw %>%
  rename(
    observation_date = observation_date,
    tlnrescons = tlnrescons
  ) %>%
  mutate(
    observation_date = ymd(observation_date)
  ) %>%
  select(observation_date, tlnrescons)

indpro_clean <- indpro_raw %>%
  rename(
    observation_date = observation_date,
    indpro = industrial_pro
  ) %>%
  mutate(
    observation_date = ymd(observation_date)
  ) %>%
  select(observation_date, indpro)

dgs10_clean <- dgs10_raw %>%
  rename(
    observation_date = observation_date,
    dgs10 = interest_rate_10y
  ) %>%
  mutate(
    observation_date = ymd(observation_date)
  ) %>%
  select(observation_date, dgs10)

ecom_clean
pnfi_clean
tlnrescons_clean
indpro_clean
dgs10_clean
```

```{r}
# CHUNK 27 — Convert Warehouse External Indicators to Quarterly Frequency
# Reuses to_quarterly_mean() defined in CHUNK 11.

ecom_q <- to_quarterly_mean(ecom_clean, "ecom_sales", "ecom_sales")
pnfi_q <- to_quarterly_mean(pnfi_clean, "pnfi", "pnfi")
tlnrescons_q <- to_quarterly_mean(tlnrescons_clean, "tlnrescons", "tlnrescons")
```

````
indpro_q <- to_quarterly_mean(indpro_clean, "indpro", "indpro")
dgs10_q <- to_quarterly_mean(dgs10_clean, "dgs10", "dgs10")

ecom_q
pnfi_q
tlnrescons_q
indpro_q
dgs10_q
```

```{r}
# CHUNK 28 — Build Warehouse Quarterly Internal Dataset

warehouse_quarterly <- baseline_quarterly_complete %>%
  filter(project_type == "WAREHOUSE") %>%
  select(quarter, labor_hours, job_starts) %>%
  arrange(quarter)

warehouse_quarterly
```

```{r}
# CHUNK 29 — Merge Warehouse Quarterly External Indicators

warehouse_external_indicators_q <- ecom_q %>%
  full_join(pnfi_q, by = "quarter") %>%
  full_join(tlnrescons_q, by = "quarter") %>%
  full_join(indpro_q, by = "quarter") %>%
  full_join(dgs10_q, by = "quarter") %>%
  arrange(quarter)

warehouse_external_indicators_q
```

```{r}
# CHUNK 30 — Quick Visual Check of Warehouse External Indicators

warehouse_external_long <- warehouse_external_indicators_q %>%
  pivot_longer(
    cols = -quarter,
    names_to = "indicator",
    values_to = "value"
  )

ggplot(warehouse_external_long, aes(x = quarter, y = value, color = indicator)) +
  geom_line(linewidth = 1) +
  facet_wrap(~ indicator, scales = "free_y", ncol = 2) +
  labs(
    title = "Quarterly Warehouse External Indicators",
````

````
    x = "Quarter",
    y = "Value",
    color = "Indicator"
  ) +
  theme_minimal(base_size = 12)
```

```{r}
# CHUNK 31 — Build Warehouse Modeling Table

warehouse_model_data <- warehouse_quarterly %>%
  left_join(ecom_q, by = "quarter") %>%
  left_join(pnfi_q, by = "quarter") %>%
  left_join(tlnrescons_q, by = "quarter") %>%
  left_join(indpro_q, by = "quarter") %>%
  left_join(dgs10_q, by = "quarter") %>%
  arrange(quarter)

glimpse(warehouse_model_data)
head(warehouse_model_data)
```

```{r}
# CHUNK 32 — Create Lagged Versions of Internal and External Variables

warehouse_model_data <- warehouse_model_data %>%
  arrange(quarter) %>%
  mutate(
    qtr = factor(
      paste0("Q", lubridate::quarter(quarter)),
      levels = c("Q1", "Q2", "Q3", "Q4")
    ),

    labor_hours_lag1 = lag(labor_hours, 1),
    labor_hours_lag2 = lag(labor_hours, 2),
    labor_hours_lag3 = lag(labor_hours, 3),
    labor_hours_lag4 = lag(labor_hours, 4),

    ecom_lag1 = lag(ecom_sales, 1),
    ecom_lag2 = lag(ecom_sales, 2),
    ecom_lag3 = lag(ecom_sales, 3),
    ecom_lag4 = lag(ecom_sales, 4),

    pnfi_lag1 = lag(pnfi, 1),
    pnfi_lag2 = lag(pnfi, 2),
    pnfi_lag3 = lag(pnfi, 3),
    pnfi_lag4 = lag(pnfi, 4),

    tlnrescons_lag1 = lag(tlnrescons, 1),
````

````
    tlnrescons_lag2 = lag(tlnrescons, 2),
    tlnrescons_lag3 = lag(tlnrescons, 3),
    tlnrescons_lag4 = lag(tlnrescons, 4),

    indpro_lag1 = lag(indpro, 1),
    indpro_lag2 = lag(indpro, 2),
    indpro_lag3 = lag(indpro, 3),
    indpro_lag4 = lag(indpro, 4),

    dgs10_lag1 = lag(dgs10, 1),
    dgs10_lag2 = lag(dgs10, 2),
    dgs10_lag3 = lag(dgs10, 3),
    dgs10_lag4 = lag(dgs10, 4)
  )

glimpse(warehouse_model_data)
```

```{r}
# CHUNK 33 — Create Horizon-Specific Forecasting Function for Warehouse

fit_warehouse_horizon_models <- function(data, horizon, test_periods = 4) {

  horizon_data <- data %>%
    transmute(
      quarter,
      qtr,
      labor_hours,
      labor_hours_lag = .data[[paste0("labor_hours_lag", horizon)]],
      ecom_lag = .data[[paste0("ecom_lag", horizon)]],
      pnfi_lag = .data[[paste0("pnfi_lag", horizon)]],
      tlnrescons_lag = .data[[paste0("tlnrescons_lag", horizon)]],
      indpro_lag = .data[[paste0("indpro_lag", horizon)]],
      dgs10_lag = .data[[paste0("dgs10_lag", horizon)]]
    ) %>%
    drop_na()

  n_obs <- nrow(horizon_data)

  if (n_obs < 8) {
    stop(paste("Not enough observations available for horizon", horizon))
  }

  test_n <- min(test_periods, max(2, floor(n_obs * 0.25)))
  train_n <- n_obs - test_n

  train_data <- horizon_data %>%
    slice(1:train_n)
````

```
  test_data <- horizon_data %>%
    slice((train_n + 1):n_obs)

  baseline_model <- lm(
    labor_hours ~ labor_hours_lag + qtr,
    data = train_data
  )

  augmented_model <- lm(
    labor_hours ~ labor_hours_lag +
      ecom_lag +
      pnfi_lag +
      tlnrescons_lag +
      indpro_lag +
      dgs10_lag +
      qtr,
    data = train_data
  )

  test_predictions <- test_data %>%
    mutate(
      baseline_forecast = predict(baseline_model, newdata = test_data),
      augmented_forecast = predict(augmented_model, newdata = test_data)
    )

  baseline_metrics <- calculate_error_metrics(
    actual = test_predictions$labor_hours,
    predicted = test_predictions$baseline_forecast
  )

  augmented_metrics <- calculate_error_metrics(
    actual = test_predictions$labor_hours,
    predicted = test_predictions$augmented_forecast
  )

  list(
    horizon = horizon,
    horizon_data = horizon_data,
    train_data = train_data,
    test_data = test_data,
    baseline_model = baseline_model,
    augmented_model = augmented_model,
    predictions = test_predictions,
    baseline_metrics = baseline_metrics,
    augmented_metrics = augmented_metrics
  )
}
```

````
```{r}
# CHUNK 34 — Run True 1Q, 2Q, 3Q, and 4Q Ahead Warehouse Models

warehouse_horizon_results <- list(
  `1Q Ahead` = fit_warehouse_horizon_models(warehouse_model_data, horizon = 1),
  `2Q Ahead` = fit_warehouse_horizon_models(warehouse_model_data, horizon = 2),
  `3Q Ahead` = fit_warehouse_horizon_models(warehouse_model_data, horizon = 3),
  `4Q Ahead` = fit_warehouse_horizon_models(warehouse_model_data, horizon = 4)
)

names(warehouse_horizon_results)
```

```{r}
# CHUNK 35 — Warehouse Forecast Error Summary Table

warehouse_forecast_error_summary <- bind_rows(
  lapply(names(warehouse_horizon_results), function(h_name) {
    result <- warehouse_horizon_results[[h_name]]

    tibble(
      segment = "Warehouse",
      forecast_horizon = h_name,
      baseline_MAE = result$baseline_metrics$MAE,
      enhanced_MAE = result$augmented_metrics$MAE,
      baseline_MAPE = result$baseline_metrics$MAPE,
      enhanced_MAPE = result$augmented_metrics$MAPE,
      baseline_RMSE = result$baseline_metrics$RMSE,
      enhanced_RMSE = result$augmented_metrics$RMSE
    )
  })
) %>%
  mutate(
    MAE_lift_pct = ((baseline_MAE - enhanced_MAE) / baseline_MAE) * 100,
    MAPE_lift_pct = ((baseline_MAPE - enhanced_MAPE) / baseline_MAPE) * 100,
    RMSE_lift_pct = ((baseline_RMSE - enhanced_RMSE) / baseline_RMSE) * 100
  )

warehouse_forecast_error_summary %>%
  mutate(
    across(
      c(
        baseline_MAE, enhanced_MAE,
        baseline_RMSE, enhanced_RMSE
      ),
      ~ comma(round(.))
    ),
    across(
      c(
````
````

````
        baseline_MAPE, enhanced_MAPE,
        MAE_lift_pct, MAPE_lift_pct, RMSE_lift_pct
      ),
      ~ percent(. / 100, accuracy = 0.1)
    )
  ) %>%
  kable(
    caption = "Warehouse Forecast Error Summary by Horizon"
  ) %>%
  kable_styling(full_width = FALSE)
```

```{r}
# CHUNK 36 — Warehouse Lift in Accuracy Summary Table

warehouse_lift_summary <- warehouse_forecast_error_summary %>%
  select(
    forecast_horizon,
    baseline_RMSE,
    enhanced_RMSE,
    RMSE_lift_pct,
    baseline_MAE,
    enhanced_MAE,
    MAE_lift_pct
  )

warehouse_lift_summary %>%
  mutate(
    baseline_RMSE = comma(round(baseline_RMSE)),
    enhanced_RMSE = comma(round(enhanced_RMSE)),
    RMSE_lift_pct = percent(RMSE_lift_pct / 100, accuracy = 0.1),
    baseline_MAE = comma(round(baseline_MAE)),
    enhanced_MAE = comma(round(enhanced_MAE)),
    MAE_lift_pct = percent(MAE_lift_pct / 100, accuracy = 0.1)
  ) %>%
  kable(
    caption = "Warehouse Lift in Accuracy from External Indicators"
  ) %>%
  kable_styling(full_width = FALSE)
```

```{r}
# CHUNK 37 — Warehouse Baseline and Augmented Coefficients by Horizon

warehouse_coefficients_by_horizon <- bind_rows(
  lapply(names(warehouse_horizon_results), function(h_name) {
    result <- warehouse_horizon_results[[h_name]]

    baseline_tidy <- tidy(result$baseline_model) %>%
````

```
   mutate(
    model_type = "Baseline",
    forecast_horizon = h_name
   )

  augmented_tidy <- tidy(result$augmented_model) %>%
   mutate(
    model_type = "Augmented",
    forecast_horizon = h_name
   )

  bind_rows(baseline_tidy, augmented_tidy)
 })
) %>%
 mutate(
  significance = case_when(
   p.value < 0.01 ~ "***",
   p.value < 0.05 ~ "**",
   p.value < 0.10 ~ "*",
   TRUE ~ ""
  )
 )

warehouse_coefficients_by_horizon %>%
 mutate(
  estimate = round(estimate, 4),
  std.error = round(std.error, 4),
  statistic = round(statistic, 4),
  p.value = round(p.value, 4)
 ) %>%
 kable(
  caption = "Warehouse Regression Coefficients by Horizon"
 ) %>%
 kable_styling(full_width = FALSE)
```

```{r}
# CHUNK 38 — Warehouse Indicators Tested Summary by Horizon

warehouse_indicator_tested_summary <- warehouse_coefficients_by_horizon %>%
 filter(
  model_type == "Augmented",
  !term %in% c("(Intercept)", "labor_hours_lag"),
  !str_detect(term, "^qtr")
 ) %>%
 mutate(
  direction = case_when(
   estimate > 0 ~ "Positive",
   estimate < 0 ~ "Negative",
```

````
      TRUE ~ "Neutral"
    ),
    included_in_final_model = if_else(p.value < 0.10, "Yes", "No")
  ) %>%
  select(
    forecast_horizon,
    indicator = term,
    estimate,
    p.value,
    significance,
    direction,
    included_in_final_model
  )

warehouse_indicator_tested_summary %>%
  mutate(
    estimate = round(estimate, 4),
    p.value = round(p.value, 4)
  ) %>%
  kable(
    caption = "Warehouse Indicators Tested by Horizon"
  ) %>%
  kable_styling(full_width = FALSE)
```

```{r}
# CHUNK 39 — Warehouse RMSE Comparison Plot by Horizon

warehouse_rmse_plot_data <- warehouse_forecast_error_summary %>%
  select(
    forecast_horizon,
    baseline_RMSE,
    enhanced_RMSE
  ) %>%
  pivot_longer(
    cols = c(baseline_RMSE, enhanced_RMSE),
    names_to = "model_type",
    values_to = "RMSE"
  ) %>%
  mutate(
    model_type = recode(
      model_type,
      baseline_RMSE = "Baseline",
      enhanced_RMSE = "Enhanced"
    ),
    forecast_horizon = factor(
      forecast_horizon,
      levels = c("1Q Ahead", "2Q Ahead", "3Q Ahead", "4Q Ahead")
    )
````

````
  )

ggplot(warehouse_rmse_plot_data, aes(x = forecast_horizon, y = RMSE, fill = model_type)) +
  geom_col(position = "dodge") +
  scale_y_continuous(labels = comma) +
  labs(
    title = "Warehouse RMSE by Forecast Horizon",
    subtitle = "Baseline vs Enhanced Models",
    x = "Forecast Horizon",
    y = "RMSE",
    fill = "Model"
  ) +
  theme_minimal(base_size = 12)
```

```{r}
# CHUNK 40 — Warehouse Actual vs Predicted Plot for 4Q Ahead

warehouse_plot_data <- warehouse_horizon_results[["4Q Ahead"]]$predictions %>%
  select(quarter, labor_hours, baseline_forecast, augmented_forecast) %>%
  pivot_longer(
    cols = c(labor_hours, baseline_forecast, augmented_forecast),
    names_to = "series",
    values_to = "value"
  ) %>%
  mutate(
    series = recode(
      series,
      labor_hours = "Actual",
      baseline_forecast = "Baseline Forecast",
      augmented_forecast = "Enhanced Forecast"
    )
  )

ggplot(warehouse_plot_data, aes(x = quarter, y = value, color = series, linetype = series)) +
  geom_line(linewidth = 1.2) +
  scale_y_continuous(labels = comma) +
  labs(
    title = "Warehouse 4Q Ahead Forecast: Actual vs Predicted",
    subtitle = "Test Period Comparison",
    x = "Quarter",
    y = "Labor Hours",
    color = "",
    linetype = ""
  ) +
  theme_minimal(base_size = 12)
```

```{r}
````

````
# CHUNK 41 — Combine Data Center and Warehouse Forecast Error Results

combined_forecast_summary <- bind_rows(
  dc_forecast_error_summary,
  warehouse_forecast_error_summary
) %>%
  select(
    segment,
    forecast_horizon,
    baseline_RMSE,
    enhanced_RMSE,
    RMSE_lift_pct,
    baseline_MAE,
    enhanced_MAE,
    MAE_lift_pct
  ) %>%
  mutate(
    forecast_horizon = factor(
      forecast_horizon,
      levels = c("1Q Ahead", "2Q Ahead", "3Q Ahead", "4Q Ahead")
    ),
    segment = factor(
      segment,
      levels = c("Data Center", "Warehouse")
    )
  )

combined_forecast_summary
```

```{r}
# CHUNK 42 — Combined RMSE Lift Table for Data Center vs Warehouse

combined_forecast_summary %>%
  mutate(
    baseline_RMSE = comma(round(baseline_RMSE)),
    enhanced_RMSE = comma(round(enhanced_RMSE)),
    RMSE_lift_pct = percent(RMSE_lift_pct / 100, accuracy = 0.1),
    baseline_MAE = comma(round(baseline_MAE)),
    enhanced_MAE = comma(round(enhanced_MAE)),
    MAE_lift_pct = percent(MAE_lift_pct / 100, accuracy = 0.1)
  ) %>%
  arrange(segment, forecast_horizon) %>%
  kable(
    caption = "Forecast Accuracy Improvement by Segment and Horizon"
  ) %>%
  kable_styling(full_width = FALSE)
```
````
````

```{r}
# CHUNK 43 — RMSE Lift Comparison Plot Across Segments and Horizons

combined_rmse_plot_data <- combined_forecast_summary %>%
  select(segment, forecast_horizon, RMSE_lift_pct)

ggplot(combined_rmse_plot_data,
       aes(x = forecast_horizon, y = RMSE_lift_pct, fill = segment)) +
  geom_col(position = "dodge") +
  geom_hline(yintercept = 0, linetype = "dashed") +
  scale_y_continuous(labels = function(x) paste0(round(x), "%")) +
  labs(
    title = "RMSE Improvement from External Indicators",
    subtitle = "Comparison Across Segments and Forecast Horizons",
    x = "Forecast Horizon",
    y = "RMSE Lift %",
    fill = "Segment"
  ) +
  theme_minimal(base_size = 12)
```

```{r}
# CHUNK 44 — Best Forecast Horizon by Segment

best_horizon_summary <- combined_forecast_summary %>%
  group_by(segment) %>%
  slice_max(RMSE_lift_pct, n = 1) %>%
  ungroup()

best_horizon_summary %>%
  mutate(
    baseline_RMSE = comma(round(baseline_RMSE)),
    enhanced_RMSE = comma(round(enhanced_RMSE)),
    RMSE_lift_pct = percent(RMSE_lift_pct / 100, accuracy = 0.1),
    baseline_MAE = comma(round(baseline_MAE)),
    enhanced_MAE = comma(round(enhanced_MAE)),
    MAE_lift_pct = percent(MAE_lift_pct / 100, accuracy = 0.1)
  ) %>%
  kable(
    caption = "Best Forecast Horizon by Segment"
  ) %>%
  kable_styling(full_width = FALSE)
```

```{r}
# CHUNK 45 — Final Business Takeaway Table

final_business_takeaway <- tibble(
  segment = c("Data Center", "Warehouse"),
```

````
  preferred_horizon = c("3Q Ahead", "4Q Ahead"),
  key_indicator_story = c(
    "Technology investment and data center construction indicators showed the strongest signal approximately 3 quarters ahead.",
    "Warehouse demand showed the strongest longer-term signal approximately 4 quarters ahead, with interest rates becoming more important at longer horizons."
  ),
  business_takeaway = c(
    "Lithko may be able to identify rising data center demand roughly 3 quarters before labor demand accelerates.",
    "Lithko may be able to identify warehouse demand shifts roughly 4 quarters in advance using external macro indicators."
  )
)

final_business_takeaway %>%
  kable(
    caption = "Final Business Takeaways by Segment"
  ) %>%
  kable_styling(full_width = FALSE)
```

```{r}
# CHUNK 46 — Business-Preferred Horizon Summary

business_preferred_horizon <- tibble(
  segment = c("Data Center", "Warehouse"),
  preferred_horizon = c("3Q Ahead", "4Q Ahead")
) %>%
  left_join(
    combined_forecast_summary,
    by = c("segment", "preferred_horizon" = "forecast_horizon")
  )

business_preferred_horizon %>%
  mutate(
    baseline_RMSE = comma(round(baseline_RMSE)),
    enhanced_RMSE = comma(round(enhanced_RMSE)),
    RMSE_lift_pct = percent(RMSE_lift_pct / 100, accuracy = 0.1),
    baseline_MAE = comma(round(baseline_MAE)),
    enhanced_MAE = comma(round(enhanced_MAE)),
    MAE_lift_pct = percent(MAE_lift_pct / 100, accuracy = 0.1)
  ) %>%
  kable(
    caption = "Business-Preferred Forecast Horizon by Segment"
  ) %>%
  kable_styling(full_width = FALSE)
```
````
````

````
```{r}
# CHUNK 47 — Combined Indicator Story Table

data_center_indicator_story <- dc_indicator_tested_summary %>%
  mutate(segment = "Data Center")

warehouse_indicator_story <- warehouse_indicator_tested_summary %>%
  mutate(segment = "Warehouse")

combined_indicator_story <- bind_rows(
  data_center_indicator_story,
  warehouse_indicator_story
) %>%
  select(
    segment,
    forecast_horizon,
    indicator,
    estimate,
    p.value,
    significance,
    direction,
    included_in_final_model
  )

combined_indicator_story %>%
  mutate(
    estimate = round(estimate, 4),
    p.value = round(p.value, 4)
  ) %>%
  kable(
    caption = "Combined Indicator Testing Summary"
  ) %>%
  kable_styling(full_width = FALSE)
```

```{r}
# CHUNK 48 — Full-Timeline Data Center Baseline Validation

dc_full_baseline_data <- dc_model_data %>%
  transmute(
    quarter,
    qtr,
    labor_hours,
    labor_hours_lag1,
    labor_hours_lag2
  ) %>%
  drop_na()

dc_full_baseline_model <- lm(
````
````

````
  labor_hours ~ labor_hours_lag1 + labor_hours_lag2 + qtr,
  data = dc_full_baseline_data
)

dc_full_baseline_data <- dc_full_baseline_data %>%
  mutate(
    baseline_fitted = predict(dc_full_baseline_model, newdata = dc_full_baseline_data)
  )

dc_full_baseline_data %>%
  select(quarter, labor_hours, baseline_fitted)
```

```{r}
# CHUNK 49 — Full-Timeline Data Center Actual vs Baseline Plot

dc_full_baseline_plot_data <- dc_full_baseline_data %>%
  select(quarter, labor_hours, baseline_fitted) %>%
  pivot_longer(
    cols = c(labor_hours, baseline_fitted),
    names_to = "series",
    values_to = "value"
  ) %>%
  mutate(
    series = recode(
      series,
      labor_hours = "Actual",
      baseline_fitted = "Baseline Fitted"
    )
  )

ggplot(dc_full_baseline_plot_data,
       aes(x = quarter, y = value, color = series, linetype = series)) +
  geom_line(linewidth = 1.2) +
  scale_y_continuous(labels = comma) +
  labs(
    title = "Data Center Full-Timeline Actual vs Baseline",
    subtitle = "Baseline validation over the full available timeline",
    x = "Quarter",
    y = "Labor Hours",
    color = "",
    linetype = ""
  ) +
  theme_minimal(base_size = 12)
```

```{r}
# CHUNK 50 — Full-Timeline Warehouse Baseline Validation
````

````
warehouse_full_baseline_data <- warehouse_model_data %>%
  transmute(
    quarter,
    qtr,
    labor_hours,
    labor_hours_lag1,
    labor_hours_lag2
  ) %>%
  drop_na()

warehouse_full_baseline_model <- lm(
  labor_hours ~ labor_hours_lag1 + labor_hours_lag2 + qtr,
  data = warehouse_full_baseline_data
)

warehouse_full_baseline_data <- warehouse_full_baseline_data %>%
  mutate(
    baseline_fitted = predict(warehouse_full_baseline_model, newdata = warehouse_full_baseline_data)
  )

warehouse_full_baseline_data %>%
  select(quarter, labor_hours, baseline_fitted)
```

```{r}
# CHUNK 51 — Full-Timeline Warehouse Actual vs Baseline Plot

warehouse_full_baseline_plot_data <- warehouse_full_baseline_data %>%
  select(quarter, labor_hours, baseline_fitted) %>%
  pivot_longer(
    cols = c(labor_hours, baseline_fitted),
    names_to = "series",
    values_to = "value"
  ) %>%
  mutate(
    series = recode(
      series,
      labor_hours = "Actual",
      baseline_fitted = "Baseline Fitted"
    )
  )

ggplot(warehouse_full_baseline_plot_data,
       aes(x = quarter, y = value, color = series, linetype = series)) +
  geom_line(linewidth = 1.2) +
  scale_y_continuous(labels = comma) +
  labs(
    title = "Warehouse Full-Timeline Actual vs Baseline",
    subtitle = "Baseline validation over the full available timeline",
````

````
    x = "Quarter",
    y = "Labor Hours",
    color = "",
    linetype = ""
  ) +
  theme_minimal(base_size = 12)
```

```{r}
# CHUNK 52 — Baseline Bias Summary Table

baseline_bias_summary <- bind_rows(
  dc_full_baseline_data %>%
    summarise(
      segment = "Data Center",
      mean_actual = mean(labor_hours, na.rm = TRUE),
      mean_baseline = mean(baseline_fitted, na.rm = TRUE),
      mean_error = mean(baseline_fitted - labor_hours, na.rm = TRUE),
      mean_abs_error = mean(abs(baseline_fitted - labor_hours), na.rm = TRUE)
    ),
  warehouse_full_baseline_data %>%
    summarise(
      segment = "Warehouse",
      mean_actual = mean(labor_hours, na.rm = TRUE),
      mean_baseline = mean(baseline_fitted, na.rm = TRUE),
      mean_error = mean(baseline_fitted - labor_hours, na.rm = TRUE),
      mean_abs_error = mean(abs(baseline_fitted - labor_hours), na.rm = TRUE)
    )
)

baseline_bias_summary %>%
  mutate(
    across(
      c(mean_actual, mean_baseline, mean_error, mean_abs_error),
      ~ comma(round(.))
    )
  ) %>%
  kable(
    caption = "Baseline Bias Check by Segment"
  ) %>%
  kable_styling(full_width = FALSE)
```

```{r}
# CHUNK 53 — Full-Timeline Data Center Actual vs Baseline vs Enhanced

dc_full_enhanced_data <- dc_model_data %>%
  transmute(
    quarter,
````

```
    qtr,
    labor_hours,
    labor_hours_lag1,
    labor_hours_lag2,
    dc_spending_lag1,
    pce_deflator_lag1,
    tcu_lag1,
    utility_production_lag1
  ) %>%
  drop_na()

dc_full_baseline_model_1q <- lm(
  labor_hours ~ labor_hours_lag1 + labor_hours_lag2 + qtr,
  data = dc_full_enhanced_data
)

dc_full_enhanced_model_1q <- lm(
  labor_hours ~ labor_hours_lag1 + labor_hours_lag2 +
    I(dc_spending_lag1 / 100000000) +
    pce_deflator_lag1 +
    tcu_lag1 +
    utility_production_lag1 +
    qtr,
  data = dc_full_enhanced_data
)

dc_full_enhanced_data <- dc_full_enhanced_data %>%
  mutate(
    baseline_fitted = predict(dc_full_baseline_model_1q, newdata = dc_full_enhanced_data),
    enhanced_fitted = predict(dc_full_enhanced_model_1q, newdata = dc_full_enhanced_data)
  )

dc_full_enhanced_plot_data <- dc_full_enhanced_data %>%
  select(quarter, labor_hours, baseline_fitted, enhanced_fitted) %>%
  pivot_longer(
    cols = c(labor_hours, baseline_fitted, enhanced_fitted),
    names_to = "series",
    values_to = "value"
  ) %>%
  mutate(
    series = recode(
      series,
      labor_hours = "Actual",
      baseline_fitted = "Baseline Fitted",
      enhanced_fitted = "Enhanced Fitted"
    )
  )

ggplot(dc_full_enhanced_plot_data,
```

````
    aes(x = quarter, y = value, color = series, linetype = series)) +
  geom_line(linewidth = 1.2) +
  scale_y_continuous(labels = comma) +
  labs(
    title = "Data Center Full-Timeline Actual vs Baseline vs Enhanced",
    subtitle = "Validation over the full available timeline",
    x = "Quarter",
    y = "Labor Hours",
    color = "",
    linetype = ""
  ) +
  theme_minimal(base_size = 12)
```

```{r}
# CHUNK 54 — Full-Timeline Warehouse Actual vs Baseline vs Enhanced

warehouse_full_enhanced_data <- warehouse_model_data %>%
  transmute(
    quarter,
    qtr,
    labor_hours,
    labor_hours_lag1,
    labor_hours_lag2,
    ecom_lag1,
    pnfi_lag1,
    tlnrescons_lag1,
    indpro_lag1,
    dgs10_lag1
  ) %>%
  drop_na()

warehouse_full_baseline_model_1q <- lm(
  labor_hours ~ labor_hours_lag1 + labor_hours_lag2 + qtr,
  data = warehouse_full_enhanced_data
)

warehouse_full_enhanced_model_1q <- lm(
  labor_hours ~ labor_hours_lag1 + labor_hours_lag2 +
    ecom_lag1 +
    pnfi_lag1 +
    tlnrescons_lag1 +
    indpro_lag1 +
    dgs10_lag1 +
    qtr,
  data = warehouse_full_enhanced_data
)

warehouse_full_enhanced_data <- warehouse_full_enhanced_data %>%
````

```
  mutate(
    baseline_fitted = predict(warehouse_full_baseline_model_1q, newdata = warehouse_full_enhanced_data),
    enhanced_fitted = predict(warehouse_full_enhanced_model_1q, newdata =
warehouse_full_enhanced_data)
  )

warehouse_full_enhanced_plot_data <- warehouse_full_enhanced_data %>%
  select(quarter, labor_hours, baseline_fitted, enhanced_fitted) %>%
  pivot_longer(
    cols = c(labor_hours, baseline_fitted, enhanced_fitted),
    names_to = "series",
    values_to = "value"
  ) %>%
  mutate(
    series = recode(
      series,
      labor_hours = "Actual",
      baseline_fitted = "Baseline Fitted",
      enhanced_fitted = "Enhanced Fitted"
    )
  )

ggplot(warehouse_full_enhanced_plot_data,
       aes(x = quarter, y = value, color = series, linetype = series)) +
  geom_line(linewidth = 1.2) +
  scale_y_continuous(labels = comma) +
  labs(
    title = "Warehouse Full-Timeline Actual vs Baseline vs Enhanced",
    subtitle = "Validation over the full available timeline",
    x = "Quarter",
    y = "Labor Hours",
    color = "",
    linetype = ""
  ) +
  theme_minimal(base_size = 12)
```

```{r}
# CHUNK 55 — Data Center Final Sector Summary Table

dc_rsq_table <- tibble(
  baseline_r_squared = summary(dc_horizon_results[["3Q Ahead"]]$baseline_model)$r.squared,
  enhanced_r_squared = summary(dc_horizon_results[["3Q Ahead"]]$augmented_model)$r.squared
)

data_center_final_summary <- tibble(
  segment = "Data Center",
  preferred_horizon = "3Q Ahead",
  baseline_rmse = dc_forecast_error_summary %>%
```

```
    filter(forecast_horizon == "3Q Ahead") %>%
    pull(baseline_RMSE),
  enhanced_rmse = dc_forecast_error_summary %>%
    filter(forecast_horizon == "3Q Ahead") %>%
    pull(enhanced_RMSE),
  rmse_lift_pct = dc_forecast_error_summary %>%
    filter(forecast_horizon == "3Q Ahead") %>%
    pull(RMSE_lift_pct),
  baseline_mae = dc_forecast_error_summary %>%
    filter(forecast_horizon == "3Q Ahead") %>%
    pull(baseline_MAE),
  enhanced_mae = dc_forecast_error_summary %>%
    filter(forecast_horizon == "3Q Ahead") %>%
    pull(enhanced_MAE),
  mae_lift_pct = dc_forecast_error_summary %>%
    filter(forecast_horizon == "3Q Ahead") %>%
    pull(MAE_lift_pct),
  baseline_r_squared = dc_rsq_table$baseline_r_squared,
  enhanced_r_squared = dc_rsq_table$enhanced_r_squared
)

data_center_final_summary %>%
  mutate(
    baseline_rmse = comma(round(baseline_rmse)),
    enhanced_rmse = comma(round(enhanced_rmse)),
    rmse_lift_pct = percent(rmse_lift_pct / 100, accuracy = 0.1),
    baseline_mae = comma(round(baseline_mae)),
    enhanced_mae = comma(round(enhanced_mae)),
    mae_lift_pct = percent(mae_lift_pct / 100, accuracy = 0.1),
    baseline_r_squared = round(baseline_r_squared, 3),
    enhanced_r_squared = round(enhanced_r_squared, 3)
  ) %>%
  kable(
    caption = "Data Center Final Sector Summary"
  ) %>%
  kable_styling(full_width = FALSE)
```

```{r}
# CHUNK 56 — Warehouse Final Sector Summary Table

warehouse_rsq_table <- tibble(
  baseline_r_squared = summary(warehouse_horizon_results[["4Q Ahead"]]$baseline_model)$r.squared,
  enhanced_r_squared = summary(warehouse_horizon_results[["4Q Ahead"]]$augmented_model)$r.squared
)

warehouse_final_summary <- tibble(
  segment = "Warehouse",
  preferred_horizon = "4Q Ahead",
```

```
  baseline_rmse = warehouse_forecast_error_summary %>%
    filter(forecast_horizon == "4Q Ahead") %>%
    pull(baseline_RMSE),
  enhanced_rmse = warehouse_forecast_error_summary %>%
    filter(forecast_horizon == "4Q Ahead") %>%
    pull(enhanced_RMSE),
  rmse_lift_pct = warehouse_forecast_error_summary %>%
    filter(forecast_horizon == "4Q Ahead") %>%
    pull(RMSE_lift_pct),
  baseline_mae = warehouse_forecast_error_summary %>%
    filter(forecast_horizon == "4Q Ahead") %>%
    pull(baseline_MAE),
  enhanced_mae = warehouse_forecast_error_summary %>%
    filter(forecast_horizon == "4Q Ahead") %>%
    pull(enhanced_MAE),
  mae_lift_pct = warehouse_forecast_error_summary %>%
    filter(forecast_horizon == "4Q Ahead") %>%
    pull(MAE_lift_pct),
  baseline_r_squared = warehouse_rsq_table$baseline_r_squared,
  enhanced_r_squared = warehouse_rsq_table$enhanced_r_squared
)

warehouse_final_summary %>%
  mutate(
    baseline_rmse = comma(round(baseline_rmse)),
    enhanced_rmse = comma(round(enhanced_rmse)),
    rmse_lift_pct = percent(rmse_lift_pct / 100, accuracy = 0.1),
    baseline_mae = comma(round(baseline_mae)),
    enhanced_mae = comma(round(enhanced_mae)),
    mae_lift_pct = percent(mae_lift_pct / 100, accuracy = 0.1),
    baseline_r_squared = round(baseline_r_squared, 3),
    enhanced_r_squared = round(enhanced_r_squared, 3)
  ) %>%
  kable(
    caption = "Warehouse Final Sector Summary"
  ) %>%
  kable_styling(full_width = FALSE)
```

````
```{r}
# CHUNK 57 — Cross-Sector Final Summary Table

cross_sector_final_summary <- bind_rows(
  data_center_final_summary,
  warehouse_final_summary
)

cross_sector_final_summary %>%
  mutate(
````
````

```
    baseline_rmse = comma(round(baseline_rmse)),
    enhanced_rmse = comma(round(enhanced_rmse)),
    rmse_lift_pct = percent(rmse_lift_pct / 100, accuracy = 0.1),
    baseline_mae = comma(round(baseline_mae)),
    enhanced_mae = comma(round(enhanced_mae)),
    mae_lift_pct = percent(mae_lift_pct / 100, accuracy = 0.1),
    baseline_r_squared = round(baseline_r_squared, 3),
    enhanced_r_squared = round(enhanced_r_squared, 3)
  ) %>%
  kable(
    caption = "Cross-Sector Final Summary"
  ) %>%
  kable_styling(full_width = FALSE)
```

```{r}
# CHUNK 58 — Cross-Sector Preferred Indicator Table

preferred_indicator_table <- tibble(
  segment = c("Data Center", "Warehouse"),
  preferred_horizon = c("3Q Ahead", "4Q Ahead"),
  key_indicator = c("dc_spending_lag", "dgs10_lag"),
  indicator_story = c(
    "Data center construction spending provided the most useful longer-horizon signal.",
    "The 10-Year Treasury Rate provided the most useful longer-horizon signal for warehouse demand."
  )
)

preferred_indicator_table %>%
  kable(
    caption = "Preferred Indicator Story by Segment"
  ) %>%
  kable_styling(full_width = FALSE)
```

```{r}
# CHUNK 59 — Data Center Seasonality Table

dc_seasonality_table <- broom::tidy(
  dc_horizon_results[["3Q Ahead"]]$augmented_model
) %>%
  filter(str_detect(term, "qtr")) %>%
  mutate(
    quarter = case_when(
      term == "qtrQ2" ~ "Q2 vs Q1",
      term == "qtrQ3" ~ "Q3 vs Q1",
      term == "qtrQ4" ~ "Q4 vs Q1"
    ),
    direction = if_else(estimate > 0, "Positive", "Negative"),
```

```
    significance = case_when(
      p.value < 0.001 ~ "***",
      p.value < 0.01 ~ "**",
      p.value < 0.05 ~ "*",
      TRUE ~ ""
    )
  ) %>%
  select(
    quarter,
    estimate,
    p.value,
    significance,
    direction
  )

dc_seasonality_table %>%
  mutate(
    estimate = comma(round(estimate)),
    p.value = round(p.value, 4)
  ) %>%
  kable(
    caption = "Data Center Seasonality Effects (Reference Quarter = Q1)"
  ) %>%
  kable_styling(full_width = FALSE)
```

```{r}
# CHUNK 60 — Warehouse Seasonality Table

warehouse_seasonality_table <- broom::tidy(
  warehouse_horizon_results[["4Q Ahead"]]$augmented_model
) %>%
  filter(str_detect(term, "qtr")) %>%
  mutate(
    quarter = case_when(
      term == "qtrQ2" ~ "Q2 vs Q1",
      term == "qtrQ3" ~ "Q3 vs Q1",
      term == "qtrQ4" ~ "Q4 vs Q1"
    ),
    direction = if_else(estimate > 0, "Positive", "Negative"),
    significance = case_when(
      p.value < 0.001 ~ "***",
      p.value < 0.01 ~ "**",
      p.value < 0.05 ~ "*",
      TRUE ~ ""
    )
  ) %>%
  select(
    quarter,
```

```
    estimate,
    p.value,
    significance,
    direction
  )

warehouse_seasonality_table %>%
  mutate(
    estimate = comma(round(estimate)),
    p.value = round(p.value, 4)
  ) %>%
  kable(
    caption = "Warehouse Seasonality Effects (Reference Quarter = Q1)"
  ) %>%
  kable_styling(full_width = FALSE)
```

```{r}
# CHUNK 61 — Cross-Sector Seasonality Summary

cross_sector_seasonality_summary <- bind_rows(
  dc_seasonality_table %>%
    mutate(segment = "Data Center"),
  warehouse_seasonality_table %>%
    mutate(segment = "Warehouse")
) %>%
  select(
    segment,
    quarter,
    estimate,
    p.value,
    significance,
    direction
  )

cross_sector_seasonality_summary %>%
  mutate(
    estimate = comma(round(estimate)),
    p.value = round(p.value, 4)
  ) %>%
  kable(
    caption = "Cross-Sector Seasonality Summary"
  ) %>%
  kable_styling(full_width = FALSE)
```

```{r}
# CHUNK 62 — Data Center Preferred Horizon Quarter-by-Quarter Error Table
```

```r
dc_preferred_error_table <- dc_horizon_results[["3Q Ahead"]]$predictions %>%
  transmute(
    quarter,
    actual_labor_hours = labor_hours,
    baseline_forecast,
    enhanced_forecast = augmented_forecast,
    baseline_abs_error = abs(labor_hours - baseline_forecast),
    enhanced_abs_error = abs(labor_hours - augmented_forecast),
    closer_model = case_when(
      enhanced_abs_error < baseline_abs_error ~ "Enhanced",
      enhanced_abs_error > baseline_abs_error ~ "Baseline",
      TRUE ~ "Tie"
    )
  )

dc_preferred_error_table %>%
  mutate(
    across(
      c(actual_labor_hours, baseline_forecast, enhanced_forecast,
        baseline_abs_error, enhanced_abs_error),
      ~ comma(round(.))
    )
  ) %>%
  kable(
    caption = "Data Center 3Q Ahead Quarter-by-Quarter Error Comparison"
  ) %>%
  kable_styling(full_width = FALSE)
```

```{r}
# CHUNK 63 — Warehouse Preferred Horizon Quarter-by-Quarter Error Table

warehouse_preferred_error_table <- warehouse_horizon_results[["4Q Ahead"]]$predictions %>%
  transmute(
    quarter,
    actual_labor_hours = labor_hours,
    baseline_forecast,
    enhanced_forecast = augmented_forecast,
    baseline_abs_error = abs(labor_hours - baseline_forecast),
    enhanced_abs_error = abs(labor_hours - augmented_forecast),
    closer_model = case_when(
      enhanced_abs_error < baseline_abs_error ~ "Enhanced",
      enhanced_abs_error > baseline_abs_error ~ "Baseline",
      TRUE ~ "Tie"
    )
  )

warehouse_preferred_error_table %>%
  mutate(
```

```
    across(
      c(actual_labor_hours, baseline_forecast, enhanced_forecast,
        baseline_abs_error, enhanced_abs_error),
      ~ comma(round(.))
    )
  ) %>%
  kable(
    caption = "Warehouse 4Q Ahead Quarter-by-Quarter Error Comparison"
  ) %>%
  kable_styling(full_width = FALSE)
```

```{r}
# CHUNK 64 — Sector Macro Variable Mapping Table

sector_macro_variable_mapping <- tibble(
  segment = c("Data Center", "Warehouse"),
  variables_tested = c(
    "dc_spending, pce_deflator, tcu, utility_production",
    "ecom_sales, pnfi, tlnrescons, indpro, dgs10"
  ),
  variables_retained_or_most_useful = c(
    "dc_spending_lag, utility_production_lag",
    "ecom_lag, dgs10_lag"
  ),
  preferred_horizon = c("3Q Ahead", "4Q Ahead")
)

sector_macro_variable_mapping %>%
  kable(
    caption = "Sector Macro Variable Mapping"
  ) %>%
  kable_styling(full_width = FALSE)
```

```{r}
# CHUNK A1 — Load Apartment External Files

permit5_raw <- read_excel("PERMIT5.xlsx", sheet = "Monthly") %>%
  clean_names()

houst5f_raw <- read_excel("HOUST5F.xlsx", sheet = "Monthly") %>%
  clean_names()

dgs10_apartment_raw <- read_excel("DGS10.xlsx", sheet = "Daily") %>%
  clean_names()

undcon5musa_raw <- read_excel("UNDCON5MUSA.xlsx", sheet = "Monthly") %>%
  clean_names()
```

````
compu5musa_raw <- read_excel("COMPU5MUSA.xlsx", sheet = "Monthly") %>%
  clean_names()
```

```{r}
# CHUNK A2 — Clean Apartment External Files

permit5_clean <- permit5_raw %>%
  rename(
    observation_date = observation_date,
    permit5 = permit5
  ) %>%
  mutate(
    observation_date = ymd(observation_date)
  ) %>%
  select(observation_date, permit5)

houst5f_clean <- houst5f_raw %>%
  rename(
    observation_date = observation_date,
    houst5f = houst5f
  ) %>%
  mutate(
    observation_date = ymd(observation_date)
  ) %>%
  select(observation_date, houst5f)

dgs10_apartment_clean <- dgs10_apartment_raw %>%
  rename(
    observation_date = observation_date,
    dgs10 = dgs10
  ) %>%
  mutate(
    observation_date = ymd(observation_date)
  ) %>%
  select(observation_date, dgs10)

undcon5musa_clean <- undcon5musa_raw %>%
  rename(
    observation_date = observation_date,
    undcon5musa = undcon5musa
  ) %>%
  mutate(
    observation_date = ymd(observation_date)
  ) %>%
  select(observation_date, undcon5musa)

compu5musa_clean <- compu5musa_raw %>%
````

````
  rename(
    observation_date = observation_date,
    compu5musa = compu5musa
  ) %>%
  mutate(
    observation_date = ymd(observation_date)
  ) %>%
  select(observation_date, compu5musa)

permit5_clean
houst5f_clean
dgs10_apartment_clean
undcon5musa_clean
compu5musa_clean
```

```{r}
# CHUNK A3 — Build Apartment Internal Quarterly Labor-Hours Data

apartment_jobs <- jobs_enriched %>%
  filter(project_type == "APARTMENT") %>%
  mutate(
    quarter = floor_date(true_start_date, "quarter")
  )

apartment_job_starts_quarterly <- apartment_jobs %>%
  filter(!is.na(quarter)) %>%
  group_by(quarter) %>%
  summarise(
    job_starts = n_distinct(job_id),
    .groups = "drop"
  )

apartment_labor_hours_quarterly <- labor %>%
  left_join(
    apartment_jobs %>% select(job_id),
    by = "job_id"
  ) %>%
  mutate(
    quarter = floor_date(work_date, "quarter")
  ) %>%
  filter(!is.na(job_id)) %>%
  group_by(quarter) %>%
  summarise(
    labor_hours = sum(hrs, na.rm = TRUE),
    .groups = "drop"
  )

apartment_quarterly <- apartment_job_starts_quarterly %>%
````

````
  full_join(apartment_labor_hours_quarterly, by = "quarter") %>%
  arrange(quarter)

apartment_quarterly
```

```{r}
# CHUNK A4 — Remove Incomplete Final Quarter and Define Common Complete-Quarter Range

apartment_quarterly <- apartment_quarterly %>%
  filter(quarter <= last_complete_quarter)

min_q_apartment <- max(
  floor_date(min(permit5_clean$observation_date, na.rm = TRUE), "quarter"),
  floor_date(min(houst5f_clean$observation_date, na.rm = TRUE), "quarter"),
  floor_date(min(dgs10_apartment_clean$observation_date, na.rm = TRUE), "quarter"),
  floor_date(min(undcon5musa_clean$observation_date, na.rm = TRUE), "quarter"),
  floor_date(min(compu5musa_clean$observation_date, na.rm = TRUE), "quarter"),
  min(apartment_quarterly$quarter, na.rm = TRUE)
)

max_q_apartment <- min(
  max(apartment_quarterly$quarter, na.rm = TRUE),
  last_complete_quarter
)

min_q_apartment
max_q_apartment
```

```{r}
# CHUNK A5 — Convert Apartment External Series to Quarterly

to_quarterly_mean <- function(df, value_col, out_name, min_q, max_q) {
  df %>%
    mutate(quarter = floor_date(observation_date, "quarter")) %>%
    group_by(quarter) %>%
    summarise(value = mean(.data[[value_col]], na.rm = TRUE), .groups = "drop") %>%
    rename(!!out_name := value) %>%
    filter(quarter >= min_q, quarter <= max_q)
}

permit5_q <- to_quarterly_mean(permit5_clean, "permit5", "permit5", min_q_apartment, max_q_apartment)
houst5f_q <- to_quarterly_mean(houst5f_clean, "houst5f", "houst5f", min_q_apartment, max_q_apartment)
dgs10_apartment_q <- to_quarterly_mean(dgs10_apartment_clean, "dgs10", "dgs10", min_q_apartment,
max_q_apartment)
undcon5musa_q <- to_quarterly_mean(undcon5musa_clean, "undcon5musa", "undcon5musa",
min_q_apartment, max_q_apartment)
````

````
compu5musa_q <- to_quarterly_mean(compu5musa_clean, "compu5musa", "compu5musa",
min_q_apartment, max_q_apartment)

permit5_q
houst5f_q
dgs10_apartment_q
undcon5musa_q
compu5musa_q
```

```{r}
# CHUNK A6 — Merge Apartment External Quarterly Data

apartment_external_indicators_q <- permit5_q %>%
  full_join(houst5f_q, by = "quarter") %>%
  full_join(dgs10_apartment_q, by = "quarter") %>%
  full_join(undcon5musa_q, by = "quarter") %>%
  full_join(compu5musa_q, by = "quarter") %>%
  arrange(quarter)

apartment_external_indicators_q
```

```{r}
# CHUNK A7 — Merge Apartment Internal + External Data

apartment_model_data <- apartment_quarterly %>%
  filter(quarter >= min_q_apartment, quarter <= max_q_apartment) %>%
  left_join(apartment_external_indicators_q, by = "quarter") %>%
  arrange(quarter)

glimpse(apartment_model_data)
head(apartment_model_data)
```

```{r}
# CHUNK A8 — Quick Visual Check of Apartment External Indicators

apartment_external_long <- apartment_external_indicators_q %>%
  pivot_longer(
    cols = -quarter,
    names_to = "indicator",
    values_to = "value"
  )

ggplot(apartment_external_long, aes(x = quarter, y = value, color = indicator)) +
  geom_line(linewidth = 1) +
  facet_wrap(~ indicator, scales = "free_y", ncol = 2) +
  labs(
````

````
    title = "Quarterly Apartment External Indicators",
    x = "Quarter",
    y = "Value",
    color = "Indicator"
  ) +
  theme_minimal(base_size = 12)
```

```{r}
# CHUNK A9 — Create Apartment Final Modeling Variables

apartment_model_data <- apartment_model_data %>%
  arrange(quarter) %>%
  mutate(
    qtr = factor(
      paste0("Q", lubridate::quarter(quarter)),
      levels = c("Q1", "Q2", "Q3", "Q4")
    ),
    time_index = row_number(),

    labor_hours_lag1 = lag(labor_hours, 1),
    labor_hours_lag2 = lag(labor_hours, 2),
    labor_hours_lag3 = lag(labor_hours, 3),
    labor_hours_lag4 = lag(labor_hours, 4),

    permit5_lag1 = lag(permit5, 1),
    permit5_lag2 = lag(permit5, 2),
    permit5_lag3 = lag(permit5, 3),
    permit5_lag4 = lag(permit5, 4),

    dgs10_lag1 = lag(dgs10, 1),
    dgs10_lag2 = lag(dgs10, 2),
    dgs10_lag3 = lag(dgs10, 3),
    dgs10_lag4 = lag(dgs10, 4),

    undcon5musa_lag1 = lag(undcon5musa, 1),
    undcon5musa_lag2 = lag(undcon5musa, 2),
    undcon5musa_lag3 = lag(undcon5musa, 3),
    undcon5musa_lag4 = lag(undcon5musa, 4)
  )

glimpse(apartment_model_data)
```

```{r}
# CHUNK A10 — Apartment Horizon-Specific Forecasting Function

fit_apartment_horizon_models <- function(data, horizon, test_periods = 4) {
````

```r
horizon_data <- data %>%
  transmute(
    quarter,
    qtr,
    time_index,
    labor_hours,
    labor_hours_lag = .data[[paste0("labor_hours_lag", horizon)]],
    permit5_lag = .data[[paste0("permit5_lag", horizon)]],
    dgs10_lag = .data[[paste0("dgs10_lag", horizon)]]
  ) %>%
  drop_na()

n_obs <- nrow(horizon_data)

if (n_obs < 8) {
  stop(paste("Not enough observations available for horizon", horizon))
}

test_n <- min(test_periods, max(2, floor(n_obs * 0.25)))
train_n <- n_obs - test_n

train_data <- horizon_data %>%
  slice(1:train_n)

test_data <- horizon_data %>%
  slice((train_n + 1):n_obs)

baseline_model <- lm(
  labor_hours ~ labor_hours_lag + time_index + qtr,
  data = train_data
)

augmented_model <- lm(
  labor_hours ~ labor_hours_lag + permit5_lag + dgs10_lag + time_index + qtr,
  data = train_data
)

test_predictions <- test_data %>%
  mutate(
    baseline_forecast = predict(baseline_model, newdata = test_data),
    augmented_forecast = predict(augmented_model, newdata = test_data)
  )

baseline_metrics <- calculate_error_metrics(
  actual = test_predictions$labor_hours,
  predicted = test_predictions$baseline_forecast
)

augmented_metrics <- calculate_error_metrics(
```

````
    actual = test_predictions$labor_hours,
    predicted = test_predictions$augmented_forecast
  )

  list(
    horizon = horizon,
    horizon_data = horizon_data,
    train_data = train_data,
    test_data = test_data,
    baseline_model = baseline_model,
    augmented_model = augmented_model,
    predictions = test_predictions,
    baseline_metrics = baseline_metrics,
    augmented_metrics = augmented_metrics
  )
}
```

```{r}
# CHUNK A11 — Run True 1Q, 2Q, 3Q, and 4Q Ahead Apartment Models

apartment_horizon_results <- list(
  `1Q Ahead` = fit_apartment_horizon_models(apartment_model_data, horizon = 1),
  `2Q Ahead` = fit_apartment_horizon_models(apartment_model_data, horizon = 2),
  `3Q Ahead` = fit_apartment_horizon_models(apartment_model_data, horizon = 3),
  `4Q Ahead` = fit_apartment_horizon_models(apartment_model_data, horizon = 4)
)

names(apartment_horizon_results)
```

```{r}
# CHUNK A12 — Apartment Forecast Error Summary Table

apartment_forecast_error_summary <- bind_rows(
  lapply(names(apartment_horizon_results), function(h_name) {
    result <- apartment_horizon_results[[h_name]]

    tibble(
      segment = "Apartment",
      forecast_horizon = h_name,
      baseline_MAE = result$baseline_metrics$MAE,
      enhanced_MAE = result$augmented_metrics$MAE,
      baseline_MAPE = result$baseline_metrics$MAPE,
      enhanced_MAPE = result$augmented_metrics$MAPE,
      baseline_RMSE = result$baseline_metrics$RMSE,
      enhanced_RMSE = result$augmented_metrics$RMSE
    )
  })
````

~~~~
) %>%
  mutate(
    MAE_lift_pct = ((baseline_MAE - enhanced_MAE) / baseline_MAE) * 100,
    MAPE_lift_pct = ((baseline_MAPE - enhanced_MAPE) / baseline_MAPE) * 100,
    RMSE_lift_pct = ((baseline_RMSE - enhanced_RMSE) / baseline_RMSE) * 100
  )

apartment_forecast_error_summary %>%
  mutate(
    across(
      c(
        baseline_MAE, enhanced_MAE,
        baseline_RMSE, enhanced_RMSE
      ),
      ~ comma(round(.))
    ),
    across(
      c(
        baseline_MAPE, enhanced_MAPE,
        MAE_lift_pct, MAPE_lift_pct, RMSE_lift_pct
      ),
      ~ percent(. / 100, accuracy = 0.1)
    )
  ) %>%
  kable(
    caption = "Apartment Forecast Error Summary by Horizon"
  ) %>%
  kable_styling(full_width = FALSE)
```

```{r}
# CHUNK A13 — Apartment Lift in Accuracy Summary Table

apartment_lift_summary <- apartment_forecast_error_summary %>%
  select(
    forecast_horizon,
    baseline_RMSE,
    enhanced_RMSE,
    RMSE_lift_pct,
    baseline_MAE,
    enhanced_MAE,
    MAE_lift_pct
  )

apartment_lift_summary %>%
  mutate(
    baseline_RMSE = comma(round(baseline_RMSE)),
    enhanced_RMSE = comma(round(enhanced_RMSE)),
    RMSE_lift_pct = percent(RMSE_lift_pct / 100, accuracy = 0.1),
~~~~

````
    baseline_MAE = comma(round(baseline_MAE)),
    enhanced_MAE = comma(round(enhanced_MAE)),
    MAE_lift_pct = percent(MAE_lift_pct / 100, accuracy = 0.1)
  ) %>%
  kable(
    caption = "Apartment Lift in Accuracy from External Indicators"
  ) %>%
  kable_styling(full_width = FALSE)
```

```{r}
# CHUNK A14 — Apartment Baseline and Augmented Coefficients by Horizon

apartment_coefficients_by_horizon <- bind_rows(
  lapply(names(apartment_horizon_results), function(h_name) {
    result <- apartment_horizon_results[[h_name]]

    baseline_tidy <- tidy(result$baseline_model) %>%
      mutate(
        model_type = "Baseline",
        forecast_horizon = h_name
      )

    augmented_tidy <- tidy(result$augmented_model) %>%
      mutate(
        model_type = "Augmented",
        forecast_horizon = h_name
      )

    bind_rows(baseline_tidy, augmented_tidy)
  })
) %>%
  mutate(
    significance = case_when(
      p.value < 0.01 ~ "***",
      p.value < 0.05 ~ "**",
      p.value < 0.10 ~ "*",
      TRUE ~ ""
    )
  )

apartment_coefficients_by_horizon %>%
  mutate(
    estimate = round(estimate, 4),
    std.error = round(std.error, 4),
    statistic = round(statistic, 4),
    p.value = round(p.value, 4)
  ) %>%
  kable(
````

````
    caption = "Apartment Regression Coefficients by Horizon"
  ) %>%
  kable_styling(full_width = FALSE)
```

```{r}
# CHUNK A15 — Apartment Indicators Tested Summary by Horizon

apartment_indicator_tested_summary <- apartment_coefficients_by_horizon %>%
  filter(
    model_type == "Augmented",
    !term %in% c("(Intercept)", "labor_hours_lag", "time_index"),
    !str_detect(term, "^qtr")
  ) %>%
  mutate(
    direction = case_when(
      estimate > 0 ~ "Positive",
      estimate < 0 ~ "Negative",
      TRUE ~ "Neutral"
    ),
    included_in_final_model = if_else(p.value < 0.10, "Yes", "No")
  ) %>%
  select(
    forecast_horizon,
    indicator = term,
    estimate,
    p.value,
    significance,
    direction,
    included_in_final_model
  )

apartment_indicator_tested_summary %>%
  mutate(
    estimate = round(estimate, 4),
    p.value = round(p.value, 4)
  ) %>%
  kable(
    caption = "Apartment Indicators Tested by Horizon"
  ) %>%
  kable_styling(full_width = FALSE)
```

```{r}
# CHUNK A16 — Apartment RMSE Comparison Plot by Horizon

apartment_rmse_plot_data <- apartment_forecast_error_summary %>%
  select(
    forecast_horizon,
````

```
    baseline_RMSE,
    enhanced_RMSE
  ) %>%
  pivot_longer(
    cols = c(baseline_RMSE, enhanced_RMSE),
    names_to = "model_type",
    values_to = "RMSE"
  ) %>%
  mutate(
    model_type = recode(
      model_type,
      baseline_RMSE = "Baseline",
      enhanced_RMSE = "Enhanced"
    ),
    forecast_horizon = factor(
      forecast_horizon,
      levels = c("1Q Ahead", "2Q Ahead", "3Q Ahead", "4Q Ahead")
    )
  )

ggplot(apartment_rmse_plot_data, aes(x = forecast_horizon, y = RMSE, fill = model_type)) +
  geom_col(position = "dodge") +
  scale_y_continuous(labels = comma) +
  labs(
    title = "Apartment RMSE by Forecast Horizon",
    subtitle = "Baseline vs Enhanced Models",
    x = "Forecast Horizon",
    y = "RMSE",
    fill = "Model"
  ) +
  theme_minimal(base_size = 12)
```

```{r}
# CHUNK A17 — Apartment Preferred Forecast Horizon for Reporting

apartment_best_horizon_summary <- apartment_forecast_error_summary %>%
  slice_max(RMSE_lift_pct, n = 1)

apartment_best_horizon_summary %>%
  mutate(
    baseline_RMSE = comma(round(baseline_RMSE)),
    enhanced_RMSE = comma(round(enhanced_RMSE)),
    RMSE_lift_pct = percent(RMSE_lift_pct / 100, accuracy = 0.1),
    baseline_MAE = comma(round(baseline_MAE)),
    enhanced_MAE = comma(round(enhanced_MAE)),
    MAE_lift_pct = percent(MAE_lift_pct / 100, accuracy = 0.1)
  ) %>%
  kable(
```

```
    caption = "Apartment Preferred Forecast Horizon for Reporting"
  ) %>%
  kable_styling(full_width = FALSE)
```

```{r}
# CHUNK A18 — Apartment Preferred Model Selection

apartment_preferred_horizon <- apartment_best_horizon_summary %>%
  pull(forecast_horizon)

apartment_final_predictions <- apartment_horizon_results[[apartment_preferred_horizon]]$predictions %>%
  mutate(
    baseline_abs_error = abs(labor_hours - baseline_forecast),
    enhanced_abs_error = abs(labor_hours - augmented_forecast),
    closer_model = case_when(
      enhanced_abs_error < baseline_abs_error ~ "Enhanced",
      baseline_abs_error < enhanced_abs_error ~ "Baseline",
      TRUE ~ "Tie"
    )
  )

apartment_final_predictions %>%
  select(
    quarter,
    labor_hours,
    baseline_forecast,
    augmented_forecast,
    baseline_abs_error,
    enhanced_abs_error,
    closer_model
  ) %>%
  mutate(
    across(
      c(
        labor_hours,
        baseline_forecast,
        augmented_forecast,
        baseline_abs_error,
        enhanced_abs_error
      ),
      ~ comma(round(.))
    )
  ) %>%
  kable(
    caption = "Apartment Quarter-by-Quarter Forecast Comparison"
  ) %>%
  kable_styling(full_width = FALSE)
```

```{r}
# CHUNK A19 — Apartment Actual vs Baseline vs Enhanced Plot

apartment_plot_data <- apartment_final_predictions %>%
  select(
    quarter,
    labor_hours,
    baseline_forecast,
    augmented_forecast
  ) %>%
  pivot_longer(
    cols = c(labor_hours, baseline_forecast, augmented_forecast),
    names_to = "series",
    values_to = "value"
  ) %>%
  mutate(
    series = recode(
      series,
      labor_hours = "Actual",
      baseline_forecast = "Baseline Forecast",
      augmented_forecast = "Enhanced Forecast"
    )
  )

ggplot(apartment_plot_data,
       aes(x = quarter, y = value, color = series, linetype = series)) +
  geom_line(linewidth = 1.2) +
  scale_y_continuous(labels = comma) +
  labs(
    title = paste0("Apartment ", apartment_preferred_horizon, " Forecast: Actual vs Predicted"),
    subtitle = "Preferred Horizon Comparison",
    x = "Quarter",
    y = "Labor Hours",
    color = "",
    linetype = ""
  ) +
  theme_minimal(base_size = 12)
```

```{r}
# CHUNK A20 — Apartment Full Timeline Fitted Plot

apartment_full_data <- apartment_model_data %>%
  transmute(
    quarter,
    qtr,
    time_index,
    labor_hours,
```

```
    labor_hours_lag = .data[[paste0("labor_hours_lag", as.integer(substr(apartment_preferred_horizon, 1, 1)))]],
    permit5_lag = .data[[paste0("permit5_lag", as.integer(substr(apartment_preferred_horizon, 1, 1)))]],
    dgs10_lag = .data[[paste0("dgs10_lag", as.integer(substr(apartment_preferred_horizon, 1, 1)))]]
  ) %>%
  drop_na()

apartment_baseline_full <- lm(
  labor_hours ~ labor_hours_lag + time_index + qtr,
  data = apartment_full_data
)

apartment_augmented_full <- lm(
  labor_hours ~ labor_hours_lag + permit5_lag + dgs10_lag + time_index + qtr,
  data = apartment_full_data
)

apartment_full_plot_data <- apartment_full_data %>%
  mutate(
    baseline_fitted = predict(apartment_baseline_full, newdata = apartment_full_data),
    augmented_fitted = predict(apartment_augmented_full, newdata = apartment_full_data)
  ) %>%
  select(quarter, labor_hours, baseline_fitted, augmented_fitted) %>%
  pivot_longer(
    cols = c(labor_hours, baseline_fitted, augmented_fitted),
    names_to = "series",
    values_to = "value"
  ) %>%
  mutate(
    series = recode(
      series,
      labor_hours = "Actual",
      baseline_fitted = "Baseline Fitted",
      augmented_fitted = "Enhanced Fitted"
    )
  )

ggplot(apartment_full_plot_data,
       aes(x = quarter, y = value, color = series, linetype = series)) +
  geom_line(linewidth = 1.1) +
  scale_y_continuous(labels = comma) +
  labs(
    title = "Apartment Full Timeline: Actual vs Baseline vs Enhanced",
    subtitle = "Historical Fitted Comparison",
    x = "Quarter",
    y = "Labor Hours",
    color = "",
    linetype = ""
  ) +
  theme_minimal(base_size = 12)
```

````
```

```{r}
# CHUNK A21 — Apartment Seasonality Effects

apartment_seasonality_summary <- tidy(
  apartment_horizon_results[[apartment_preferred_horizon]]$baseline_model
) %>%
  filter(str_detect(term, "^qtr")) %>%
  mutate(
    quarter = recode(
      term,
      qtrQ2 = "Q2 vs Q1",
      qtrQ3 = "Q3 vs Q1",
      qtrQ4 = "Q4 vs Q1"
    ),
    significance = case_when(
      p.value < 0.01 ~ "***",
      p.value < 0.05 ~ "**",
      p.value < 0.10 ~ "*",
      TRUE ~ ""
    ),
    direction = case_when(
      estimate > 0 ~ "Positive",
      estimate < 0 ~ "Negative",
      TRUE ~ "Neutral"
    )
  ) %>%
  select(
    quarter,
    estimate,
    p.value,
    significance,
    direction
  )

apartment_seasonality_summary %>%
  mutate(
    estimate = comma(round(estimate)),
    p.value = round(p.value, 4)
  ) %>%
  kable(
    caption = "Apartment Seasonality Effects (Reference Quarter = Q1)"
  ) %>%
  kable_styling(full_width = FALSE)
```

```{r}
# CHUNK A22 — Apartment Final Indicator Story
````

````
apartment_indicator_story <- tibble(
  segment = "Apartment",
  preferred_horizon = apartment_preferred_horizon,
  variables_tested = "permit5, houst5f, dgs10, undcon5musa, compu5musa",
  variables_retained_or_most_useful = "permit5, dgs10, time_index",
  key_story = "Apartment permit activity provided the clearest external directional signal, but no tested external indicator combination produced a reliable 3Q–4Q forecasting improvement over the baseline.",
  business_takeaway = "For Apartment, the internal baseline model remained stronger than the enhanced models. This suggests apartment demand is driven more by internal history, trend, and seasonality than by the external macro indicators tested."
)

apartment_indicator_story %>%
  kable(
    caption = "Apartment Final Indicator Story"
  ) %>%
  kable_styling(full_width = FALSE)

apartment_indicator_story <- tibble(
  segment = "Apartment",
  preferred_horizon = apartment_preferred_horizon,
  variables_tested = "permit5, houst5f, dgs10, undcon5musa, compu5musa",
  variables_retained_or_most_useful = "permit5, dgs10, time_index",
  key_story = "Apartment permit activity provided the clearest external directional signal, but no tested external indicator combination produced a reliable 3Q–4Q forecasting improvement over the baseline.",
  business_takeaway = "For Apartment, the internal baseline model remained stronger than the enhanced models. This suggests apartment demand is driven more by internal history, trend, and seasonality than by the external macro indicators tested."
)
```

```{r}
# CHUNK A23 — Apartment Final Sector Summary

apartment_final_summary <- tibble(
  segment = "Apartment",
  preferred_horizon = apartment_preferred_horizon,
  baseline_rmse = apartment_forecast_error_summary %>%
    filter(forecast_horizon == apartment_preferred_horizon) %>%
    pull(baseline_RMSE),
  enhanced_rmse = apartment_forecast_error_summary %>%
    filter(forecast_horizon == apartment_preferred_horizon) %>%
    pull(enhanced_RMSE),
  rmse_lift_pct = apartment_forecast_error_summary %>%
    filter(forecast_horizon == apartment_preferred_horizon) %>%
    pull(RMSE_lift_pct),
  baseline_mae = apartment_forecast_error_summary %>%
    filter(forecast_horizon == apartment_preferred_horizon) %>%
````

```
    pull(baseline_MAE),
  enhanced_mae = apartment_forecast_error_summary %>%
    filter(forecast_horizon == apartment_preferred_horizon) %>%
    pull(enhanced_MAE),
  mae_lift_pct = apartment_forecast_error_summary %>%
    filter(forecast_horizon == apartment_preferred_horizon) %>%
    pull(MAE_lift_pct),
  baseline_r_squared =
summary(apartment_horizon_results[[apartment_preferred_horizon]]$baseline_model)$adj.r.squared,
  enhanced_r_squared =
summary(apartment_horizon_results[[apartment_preferred_horizon]]$augmented_model)$adj.r.squared
)

apartment_final_summary %>%
  mutate(
    baseline_rmse = comma(round(baseline_rmse)),
    enhanced_rmse = comma(round(enhanced_rmse)),
    rmse_lift_pct = percent(rmse_lift_pct / 100, accuracy = 0.1),
    baseline_mae = comma(round(baseline_mae)),
    enhanced_mae = comma(round(enhanced_mae)),
    mae_lift_pct = percent(mae_lift_pct / 100, accuracy = 0.1),
    baseline_r_squared = round(baseline_r_squared, 3),
    enhanced_r_squared = round(enhanced_r_squared, 3)
  ) %>%
  kable(
    caption = "Apartment Final Sector Summary"
  ) %>%
  kable_styling(full_width = FALSE)
```

```{r}
# CHUNK A24 — Apartment Long-Horizon Focused Models (3Q & 4Q Only)

fit_apartment_long_horizon <- function(data, horizon, vars) {

  model_data <- data %>%
    transmute(
      quarter,
      qtr,
      labor_hours,
      labor_hours_lag = .data[[paste0("labor_hours_lag", horizon)]],
      permit5_lag = .data[[paste0("permit5_lag", horizon)]],
      dgs10_lag = .data[[paste0("dgs10_lag", horizon)]],
      undcon5musa_lag = .data[[paste0("undcon5musa_lag", horizon)]]
    ) %>%
    drop_na()

  n_obs <- nrow(model_data)
```

```
  test_n <- min(4, max(2, floor(n_obs * 0.25)))
  train_n <- n_obs - test_n

  train_data <- model_data %>% slice(1:train_n)
  test_data  <- model_data %>% slice((train_n + 1):n_obs)

  # Baseline (no external indicators)
  baseline_model <- lm(
    labor_hours ~ labor_hours_lag + qtr,
    data = train_data
  )

  # Build formula dynamically
  formula_text <- paste(
    "labor_hours ~ labor_hours_lag +",
    paste(vars, collapse = " + "),
    "+ qtr"
  )

  enhanced_model <- lm(
    as.formula(formula_text),
    data = train_data
  )

  test_data <- test_data %>%
    mutate(
      baseline_pred = predict(baseline_model, newdata = test_data),
      enhanced_pred = predict(enhanced_model, newdata = test_data)
    )

  baseline_metrics <- calculate_error_metrics(
    test_data$labor_hours,
    test_data$baseline_pred
  )

  enhanced_metrics <- calculate_error_metrics(
    test_data$labor_hours,
    test_data$enhanced_pred
  )

  tibble(
    horizon = paste0(horizon, "Q Ahead"),
    model = paste(vars, collapse = " + "),
    baseline_RMSE = baseline_metrics$RMSE,
    enhanced_RMSE = enhanced_metrics$RMSE,
    RMSE_lift_pct = ((baseline_metrics$RMSE - enhanced_metrics$RMSE) / baseline_metrics$RMSE) * 100
  )
}
```

```{r}
# CHUNK A25 — Test Long-Horizon Indicator Combinations

apartment_long_horizon_results <- bind_rows(

  # 3Q models
  fit_apartment_long_horizon(apartment_model_data, 3,
                             vars = c("permit5_lag")),
  fit_apartment_long_horizon(apartment_model_data, 3,
                             vars = c("permit5_lag", "undcon5musa_lag")),
  fit_apartment_long_horizon(apartment_model_data, 3,
                             vars = c("permit5_lag", "dgs10_lag")),
  fit_apartment_long_horizon(apartment_model_data, 3,
                             vars = c("permit5_lag", "undcon5musa_lag", "dgs10_lag")),

  # 4Q models
  fit_apartment_long_horizon(apartment_model_data, 4,
                             vars = c("permit5_lag")),
  fit_apartment_long_horizon(apartment_model_data, 4,
                             vars = c("permit5_lag", "undcon5musa_lag")),
  fit_apartment_long_horizon(apartment_model_data, 4,
                             vars = c("permit5_lag", "dgs10_lag")),
  fit_apartment_long_horizon(apartment_model_data, 4,
                             vars = c("permit5_lag", "undcon5musa_lag", "dgs10_lag"))
)

apartment_long_horizon_results %>%
  mutate(
    baseline_RMSE = comma(round(baseline_RMSE)),
    enhanced_RMSE = comma(round(enhanced_RMSE)),
    RMSE_lift_pct = percent(RMSE_lift_pct / 100, accuracy = 0.1)
  ) %>%
  arrange(desc(RMSE_lift_pct)) %>%
  kable(
    caption = "Apartment Long-Horizon (3Q–4Q) Model Comparison"
  ) %>%
  kable_styling(full_width = FALSE)
```

```{r}
# CHUNK O1 — Load Office External Files

uspbs_raw <- read_excel("USPBS.xlsx", sheet = "Monthly") %>% clean_names()
tlnrescons_raw <- read_excel("TLNRESCONS.xlsx", sheet = "Monthly") %>% clean_names()
dgs10_office_raw <- read_excel("DGS10.xlsx", sheet = "Daily") %>% clean_names()
pnfi_raw <- read_excel("PNFI.xlsx", sheet = "Quarterly") %>% clean_names()
```

```{r}
# CHUNK O2 — Clean Office External Files

uspbs_clean <- uspbs_raw %>%
  mutate(observation_date = ymd(observation_date)) %>%
  select(observation_date, uspbs)

tlnrescons_clean <- tlnrescons_raw %>%
  mutate(observation_date = ymd(observation_date)) %>%
  select(observation_date, tlnrescons)

dgs10_office_clean <- dgs10_office_raw %>%
  mutate(observation_date = ymd(observation_date)) %>%
  select(observation_date, dgs10)

pnfi_clean <- pnfi_raw %>%
  mutate(observation_date = ymd(observation_date)) %>%
  select(observation_date, pnfi)
```

```{r}
# CHUNK O3 — Build Office Internal Data

office_jobs <- jobs_enriched %>%
  filter(project_type == "OFFICE") %>%
  mutate(quarter = floor_date(true_start_date, "quarter"))

office_job_starts_q <- office_jobs %>%
  filter(!is.na(quarter)) %>%
  group_by(quarter) %>%
  summarise(
    job_starts = n_distinct(job_id),
    .groups = "drop"
  )

office_labor_q <- labor %>%
  semi_join(
    office_jobs %>% distinct(job_id),
    by = "job_id"
  ) %>%
  mutate(
    quarter = floor_date(work_date, "quarter")
  ) %>%
  filter(!is.na(quarter)) %>%
  group_by(quarter) %>%
  summarise(
    labor_hours = sum(hrs, na.rm = TRUE),
    .groups = "drop"
  )
```

````
office_quarterly <- office_job_starts_q %>%
  full_join(office_labor_q, by = "quarter") %>%
  arrange(quarter)

office_quarterly
```

```{r}
# CHUNK O4 — Define Time Range

office_quarterly <- office_quarterly %>%
  filter(quarter <= last_complete_quarter)

min_q_office <- max(
  floor_date(min(uspbs_clean$observation_date), "quarter"),
  floor_date(min(tlnrescons_clean$observation_date), "quarter"),
  floor_date(min(dgs10_office_clean$observation_date), "quarter"),
  floor_date(min(pnfi_clean$observation_date), "quarter"),
  min(office_quarterly$quarter)
)

max_q_office <- min(
  max(office_quarterly$quarter),
  last_complete_quarter
)
```

```{r}
# CHUNK O5 — Convert to Quarterly

to_q <- function(df, col, name) {
  df %>%
    mutate(quarter = floor_date(observation_date, "quarter")) %>%
    group_by(quarter) %>%
    summarise(value = mean(.data[[col]], na.rm = TRUE), .groups = "drop") %>%
    rename(!!name := value) %>%
    filter(quarter >= min_q_office, quarter <= max_q_office)
}

uspbs_q <- to_q(uspbs_clean, "uspbs", "uspbs")
tlnrescons_q <- to_q(tlnrescons_clean, "tlnrescons", "tlnrescons")
dgs10_office_q <- to_q(dgs10_office_clean, "dgs10", "dgs10")
pnfi_q <- to_q(pnfi_clean, "pnfi", "pnfi")
```

```{r}
# CHUNK O6 — Merge Office External Indicators
````

````
office_external_q <- uspbs_q %>%
  full_join(tlnrescons_q, by = "quarter") %>%
  full_join(dgs10_office_q, by = "quarter") %>%
  full_join(pnfi_q, by = "quarter") %>%
  arrange(quarter)
```

```{r}
# CHUNK O7 — Final Office Model Data

office_model_data <- office_quarterly %>%
  filter(quarter >= min_q_office, quarter <= max_q_office) %>%
  left_join(office_external_q, by = "quarter") %>%
  arrange(quarter)

glimpse(office_model_data)
```

```{r}
# CHUNK O8 — Create Office Lag Variables + Time Index

office_model_data <- office_model_data %>%
  arrange(quarter) %>%
  mutate(
    qtr = factor(
      paste0("Q", lubridate::quarter(quarter)),
      levels = c("Q1", "Q2", "Q3", "Q4")
    ),
    time_index = row_number(),

    labor_hours_lag1 = lag(labor_hours, 1),
    labor_hours_lag2 = lag(labor_hours, 2),
    labor_hours_lag3 = lag(labor_hours, 3),
    labor_hours_lag4 = lag(labor_hours, 4),

    uspbs_lag1 = lag(uspbs, 1),
    uspbs_lag2 = lag(uspbs, 2),
    uspbs_lag3 = lag(uspbs, 3),
    uspbs_lag4 = lag(uspbs, 4),

    tlnrescons_lag1 = lag(tlnrescons, 1),
    tlnrescons_lag2 = lag(tlnrescons, 2),
    tlnrescons_lag3 = lag(tlnrescons, 3),
    tlnrescons_lag4 = lag(tlnrescons, 4),

    dgs10_lag1 = lag(dgs10, 1),
    dgs10_lag2 = lag(dgs10, 2),
    dgs10_lag3 = lag(dgs10, 3),
    dgs10_lag4 = lag(dgs10, 4),
````

~~~~
    pnfi_lag1 = lag(pnfi, 1),
    pnfi_lag2 = lag(pnfi, 2),
    pnfi_lag3 = lag(pnfi, 3),
    pnfi_lag4 = lag(pnfi, 4)
  )

glimpse(office_model_data)
```

```{r}
# CHUNK O9 — Updated Office Horizon-Specific Forecasting Function

fit_office_horizon_models <- function(data, horizon, test_periods = 4) {

  horizon_data <- data %>%
    transmute(
      quarter,
      qtr,
      time_index,
      labor_hours,
      labor_hours_lag = .data[[paste0("labor_hours_lag", horizon)]],
      uspbs_lag = .data[[paste0("uspbs_lag", horizon)]],
      tlnrescons_lag = .data[[paste0("tlnrescons_lag", horizon)]],
      dgs10_lag = .data[[paste0("dgs10_lag", horizon)]],
      pnfi_lag = .data[[paste0("pnfi_lag", horizon)]]
    ) %>%
    drop_na()

  n_obs <- nrow(horizon_data)

  if (n_obs < 8) {
    stop(paste("Not enough observations available for horizon", horizon))
  }

  test_n <- min(test_periods, max(2, floor(n_obs * 0.25)))
  train_n <- n_obs - test_n

  train_data <- horizon_data %>% slice(1:train_n)
  test_data  <- horizon_data %>% slice((train_n + 1):n_obs)

  baseline_model <- lm(
    labor_hours ~ labor_hours_lag + time_index + qtr,
    data = train_data
  )

  augmented_model <- lm(
    labor_hours ~ labor_hours_lag +
      uspbs_lag +
~~~~

````
      tlnrescons_lag +
      dgs10_lag +
      pnfi_lag +
      time_index +
      qtr,
    data = train_data
  )

  test_predictions <- test_data %>%
    mutate(
      baseline_forecast = predict(baseline_model, newdata = test_data),
      augmented_forecast = predict(augmented_model, newdata = test_data)
    )

  baseline_metrics <- calculate_error_metrics(
    actual = test_predictions$labor_hours,
    predicted = test_predictions$baseline_forecast
  )

  augmented_metrics <- calculate_error_metrics(
    actual = test_predictions$labor_hours,
    predicted = test_predictions$augmented_forecast
  )

  list(
    horizon = horizon,
    horizon_data = horizon_data,
    train_data = train_data,
    test_data = test_data,
    baseline_model = baseline_model,
    augmented_model = augmented_model,
    predictions = test_predictions,
    baseline_metrics = baseline_metrics,
    augmented_metrics = augmented_metrics
  )
}
```

```{r}
# CHUNK O10 — Run 1Q, 2Q, 3Q, and 4Q Ahead Office Models

office_horizon_results <- list(
  `1Q Ahead` = fit_office_horizon_models(office_model_data, horizon = 1),
  `2Q Ahead` = fit_office_horizon_models(office_model_data, horizon = 2),
  `3Q Ahead` = fit_office_horizon_models(office_model_data, horizon = 3),
  `4Q Ahead` = fit_office_horizon_models(office_model_data, horizon = 4)
)

names(office_horizon_results)
````

````
```

```{r}
# CHUNK O11 — Office Forecast Error Summary

office_forecast_error_summary <- bind_rows(
  lapply(names(office_horizon_results), function(h_name) {
    result <- office_horizon_results[[h_name]]

    tibble(
      segment = "Office",
      forecast_horizon = h_name,
      baseline_MAE = result$baseline_metrics$MAE,
      enhanced_MAE = result$augmented_metrics$MAE,
      baseline_MAPE = result$baseline_metrics$MAPE,
      enhanced_MAPE = result$augmented_metrics$MAPE,
      baseline_RMSE = result$baseline_metrics$RMSE,
      enhanced_RMSE = result$augmented_metrics$RMSE
    )
  })
) %>%
  mutate(
    MAE_lift_pct = ((baseline_MAE - enhanced_MAE) / baseline_MAE) * 100,
    MAPE_lift_pct = ((baseline_MAPE - enhanced_MAPE) / baseline_MAPE) * 100,
    RMSE_lift_pct = ((baseline_RMSE - enhanced_RMSE) / baseline_RMSE) * 100
  )

office_forecast_error_summary %>%
  mutate(
    across(
      c(baseline_MAE, enhanced_MAE, baseline_RMSE, enhanced_RMSE),
      ~ comma(round(.))
    ),
    across(
      c(baseline_MAPE, enhanced_MAPE, MAE_lift_pct, MAPE_lift_pct, RMSE_lift_pct),
      ~ percent(. / 100, accuracy = 0.1)
    )
  ) %>%
  kable(
    caption = "Office Forecast Error Summary by Horizon"
  ) %>%
  kable_styling(full_width = FALSE)
```

```{r}
# CHUNK O12 — Office RMSE Comparison Plot by Horizon

office_rmse_plot_data <- office_forecast_error_summary %>%
  select(forecast_horizon, baseline_RMSE, enhanced_RMSE) %>%
````

````
  pivot_longer(
    cols = c(baseline_RMSE, enhanced_RMSE),
    names_to = "model_type",
    values_to = "RMSE"
  ) %>%
  mutate(
    model_type = recode(
      model_type,
      baseline_RMSE = "Baseline",
      enhanced_RMSE = "Enhanced"
    ),
    forecast_horizon = factor(
      forecast_horizon,
      levels = c("1Q Ahead", "2Q Ahead", "3Q Ahead", "4Q Ahead")
    )
  )

ggplot(office_rmse_plot_data, aes(x = forecast_horizon, y = RMSE, fill = model_type)) +
  geom_col(position = "dodge") +
  scale_y_continuous(labels = comma) +
  labs(
    title = "Office RMSE by Forecast Horizon",
    subtitle = "Baseline vs Enhanced Models",
    x = "Forecast Horizon",
    y = "RMSE",
    fill = "Model"
  ) +
  theme_minimal(base_size = 12)
```

```{r}
# CHUNK O13 — Office Regression Coefficients by Horizon

office_coefficients_by_horizon <- bind_rows(
  lapply(names(office_horizon_results), function(h_name) {
    result <- office_horizon_results[[h_name]]

    baseline_tidy <- tidy(result$baseline_model) %>%
      mutate(
        model_type = "Baseline",
        forecast_horizon = h_name
      )

    augmented_tidy <- tidy(result$augmented_model) %>%
      mutate(
        model_type = "Augmented",
        forecast_horizon = h_name
      )
````

````
    bind_rows(baseline_tidy, augmented_tidy)
  })
) %>%
  mutate(
    significance = case_when(
      p.value < 0.01 ~ "***",
      p.value < 0.05 ~ "**",
      p.value < 0.10 ~ "*",
      TRUE ~ ""
    )
  )

office_coefficients_by_horizon %>%
  mutate(
    estimate = round(estimate, 4),
    std.error = round(std.error, 4),
    statistic = round(statistic, 4),
    p.value = round(p.value, 4)
  ) %>%
  kable(
    caption = "Office Regression Coefficients by Horizon"
  ) %>%
  kable_styling(full_width = FALSE)
```

```{r}
# CHUNK O14 — Office Indicators Tested Summary by Horizon

office_indicator_tested_summary <- office_coefficients_by_horizon %>%
  filter(
    model_type == "Augmented",
    !term %in% c("(Intercept)", "labor_hours_lag", "time_index"),
    !str_detect(term, "^qtr")
  ) %>%
  mutate(
    direction = case_when(
      estimate > 0 ~ "Positive",
      estimate < 0 ~ "Negative",
      TRUE ~ "Neutral"
    ),
    included_in_final_model = if_else(p.value < 0.10, "Yes", "No")
  ) %>%
  select(
    forecast_horizon,
    indicator = term,
    estimate,
    p.value,
    significance,
    direction,
````

````
    included_in_final_model
  )

office_indicator_tested_summary %>%
  mutate(
    estimate = round(estimate, 4),
    p.value = round(p.value, 4)
  ) %>%
  kable(
    caption = "Office Indicators Tested by Horizon"
  ) %>%
  kable_styling(full_width = FALSE)
```

```{r}
# CHUNK O15 — Office Preferred Forecast Horizon

office_best_horizon_summary <- office_forecast_error_summary %>%
  slice_max(RMSE_lift_pct, n = 1)

office_best_horizon_summary %>%
  mutate(
    baseline_RMSE = comma(round(baseline_RMSE)),
    enhanced_RMSE = comma(round(enhanced_RMSE)),
    RMSE_lift_pct = percent(RMSE_lift_pct / 100, accuracy = 0.1),
    baseline_MAE = comma(round(baseline_MAE)),
    enhanced_MAE = comma(round(enhanced_MAE)),
    MAE_lift_pct = percent(MAE_lift_pct / 100, accuracy = 0.1)
  ) %>%
  kable(
    caption = "Office Preferred Forecast Horizon"
  ) %>%
  kable_styling(full_width = FALSE)
```

```{r}
# CHUNK O16 — Office Quarter-by-Quarter Forecast Comparison

office_preferred_horizon <- office_best_horizon_summary %>%
  pull(forecast_horizon)

office_final_predictions <- office_horizon_results[[office_preferred_horizon]]$predictions %>%
  mutate(
    baseline_abs_error = abs(labor_hours - baseline_forecast),
    enhanced_abs_error = abs(labor_hours - augmented_forecast),
    closer_model = case_when(
      enhanced_abs_error < baseline_abs_error ~ "Enhanced",
      baseline_abs_error < enhanced_abs_error ~ "Baseline",
      TRUE ~ "Tie"
````

```
    )
  )

office_final_predictions %>%
  select(
    quarter,
    labor_hours,
    baseline_forecast,
    augmented_forecast,
    baseline_abs_error,
    enhanced_abs_error,
    closer_model
  ) %>%
  mutate(
    across(
      c(labor_hours, baseline_forecast, augmented_forecast, baseline_abs_error, enhanced_abs_error),
      ~ comma(round(.))
    )
  ) %>%
  kable(
    caption = "Office Quarter-by-Quarter Forecast Comparison"
  ) %>%
  kable_styling(full_width = FALSE)
```

```{r}
# CHUNK O17 — Office Actual vs Baseline vs Enhanced Plot

office_plot_data <- office_final_predictions %>%
  select(quarter, labor_hours, baseline_forecast, augmented_forecast) %>%
  pivot_longer(
    cols = c(labor_hours, baseline_forecast, augmented_forecast),
    names_to = "series",
    values_to = "value"
  ) %>%
  mutate(
    series = recode(
      series,
      labor_hours = "Actual",
      baseline_forecast = "Baseline Forecast",
      augmented_forecast = "Enhanced Forecast"
    )
  )

ggplot(office_plot_data,
       aes(x = quarter, y = value, color = series, linetype = series)) +
  geom_line(linewidth = 1.2) +
  scale_y_continuous(labels = comma) +
  labs(
```

````
    title = paste0("Office ", office_preferred_horizon, " Forecast: Actual vs Predicted"),
    subtitle = "Preferred Horizon Comparison",
    x = "Quarter",
    y = "Labor Hours",
    color = "",
    linetype = ""
  ) +
  theme_minimal(base_size = 12)
```

```{r}
# CHUNK O18 — Office Full Timeline Fitted Plot

office_h_num <- as.integer(substr(office_preferred_horizon, 1, 1))

office_full_data <- office_model_data %>%
  transmute(
    quarter,
    qtr,
    time_index,
    labor_hours,
    labor_hours_lag = .data[[paste0("labor_hours_lag", office_h_num)]],
    uspbs_lag = .data[[paste0("uspbs_lag", office_h_num)]],
    tlnrescons_lag = .data[[paste0("tlnrescons_lag", office_h_num)]],
    dgs10_lag = .data[[paste0("dgs10_lag", office_h_num)]],
    pnfi_lag = .data[[paste0("pnfi_lag", office_h_num)]]
  ) %>%
  drop_na()

office_baseline_full <- lm(
  labor_hours ~ labor_hours_lag + time_index + qtr,
  data = office_full_data
)

office_augmented_full <- lm(
  labor_hours ~ labor_hours_lag + uspbs_lag + tlnrescons_lag + dgs10_lag + pnfi_lag + time_index + qtr,
  data = office_full_data
)

office_full_plot_data <- office_full_data %>%
  mutate(
    baseline_fitted = predict(office_baseline_full, newdata = office_full_data),
    augmented_fitted = predict(office_augmented_full, newdata = office_full_data)
  ) %>%
  select(quarter, labor_hours, baseline_fitted, augmented_fitted) %>%
  pivot_longer(
    cols = c(labor_hours, baseline_fitted, augmented_fitted),
    names_to = "series",
    values_to = "value"
````

````
  ) %>%
  mutate(
    series = recode(
      series,
      labor_hours = "Actual",
      baseline_fitted = "Baseline Fitted",
      augmented_fitted = "Enhanced Fitted"
    )
  )

ggplot(office_full_plot_data,
       aes(x = quarter, y = value, color = series, linetype = series)) +
  geom_line(linewidth = 1.1) +
  scale_y_continuous(labels = comma) +
  labs(
    title = "Office Full Timeline: Actual vs Baseline vs Enhanced",
    subtitle = "Historical Fitted Comparison",
    x = "Quarter",
    y = "Labor Hours",
    color = "",
    linetype = ""
  ) +
  theme_minimal(base_size = 12)
```

```{r}
# CHUNK O19 — Office Seasonality Effects

office_seasonality_summary <- tidy(
  office_horizon_results[[office_preferred_horizon]]$baseline_model
) %>%
  filter(str_detect(term, "^qtr")) %>%
  mutate(
    quarter = recode(
      term,
      qtrQ2 = "Q2 vs Q1",
      qtrQ3 = "Q3 vs Q1",
      qtrQ4 = "Q4 vs Q1"
    ),
    significance = case_when(
      p.value < 0.01 ~ "***",
      p.value < 0.05 ~ "**",
      p.value < 0.10 ~ "*",
      TRUE ~ ""
    ),
    direction = case_when(
      estimate > 0 ~ "Positive",
      estimate < 0 ~ "Negative",
      TRUE ~ "Neutral"
````

````
    )
  ) %>%
  select(quarter, estimate, p.value, significance, direction)

office_seasonality_summary %>%
  mutate(
    estimate = comma(round(estimate)),
    p.value = round(p.value, 4)
  ) %>%
  kable(
    caption = "Office Seasonality Effects (Reference Quarter = Q1)"
  ) %>%
  kable_styling(full_width = FALSE)
```

```{r}
# CHUNK O20 — Office Final Indicator Story

office_indicator_story <- tibble(
  segment = "Office",
  preferred_horizon = office_preferred_horizon,
  variables_tested = "uspbs, tlnrescons, dgs10, pnfi",
  variables_retained_or_most_useful = "uspbs (strongest), tlnrescons (cycle support), dgs10 and pnfi (supporting)",
  key_story = "Office-related spending showed the clearest positive signal, and the broader macro specification performed best at longer horizons, especially at 4Q ahead.",
  business_takeaway = "Office demand appears to benefit from macroeconomic signal more at longer horizons than at immediate short horizons, suggesting useful early warning potential around 4 quarters ahead."
)

office_indicator_story %>%
  kable(
    caption = "Office Final Indicator Story"
  ) %>%
  kable_styling(full_width = FALSE)
```

```{r}
# CHUNK O21 — Office Final Sector Summary

office_final_summary <- tibble(
  segment = "Office",
  preferred_horizon = office_preferred_horizon,
  baseline_rmse = office_forecast_error_summary %>%
    filter(forecast_horizon == office_preferred_horizon) %>%
    pull(baseline_RMSE),
  enhanced_rmse = office_forecast_error_summary %>%
    filter(forecast_horizon == office_preferred_horizon) %>%
    pull(enhanced_RMSE),
````

````
  rmse_lift_pct = office_forecast_error_summary %>%
    filter(forecast_horizon == office_preferred_horizon) %>%
    pull(RMSE_lift_pct),
  baseline_mae = office_forecast_error_summary %>%
    filter(forecast_horizon == office_preferred_horizon) %>%
    pull(baseline_MAE),
  enhanced_mae = office_forecast_error_summary %>%
    filter(forecast_horizon == office_preferred_horizon) %>%
    pull(enhanced_MAE),
  mae_lift_pct = office_forecast_error_summary %>%
    filter(forecast_horizon == office_preferred_horizon) %>%
    pull(MAE_lift_pct),
  baseline_r_squared =
summary(office_horizon_results[[office_preferred_horizon]]$baseline_model)$adj.r.squared,
  enhanced_r_squared =
summary(office_horizon_results[[office_preferred_horizon]]$augmented_model)$adj.r.squared
)

office_final_summary %>%
  mutate(
    baseline_rmse = comma(round(baseline_rmse)),
    enhanced_rmse = comma(round(enhanced_rmse)),
    rmse_lift_pct = percent(rmse_lift_pct / 100, accuracy = 0.1),
    baseline_mae = comma(round(baseline_mae)),
    enhanced_mae = comma(round(enhanced_mae)),
    mae_lift_pct = percent(mae_lift_pct / 100, accuracy = 0.1),
    baseline_r_squared = round(baseline_r_squared, 3),
    enhanced_r_squared = round(enhanced_r_squared, 3)
  ) %>%
  kable(
    caption = "Office Final Sector Summary"
  ) %>%
  kable_styling(full_width = FALSE)
```

```{r}
# CHUNK M1 — Load Manufacturing External Files

indpro_raw <- read_excel("INDPRO.xlsx", sheet = "Monthly") %>% clean_names()
tcu_raw <- read_excel("TCU.xlsx", sheet = "Monthly") %>% clean_names()
dgs10_man_raw <- read_excel("DGS10.xlsx", sheet = "Daily") %>% clean_names()
pnfi_man_raw <- read_excel("PNFI.xlsx", sheet = "Quarterly") %>% clean_names()
```

```{r}
# CHUNK M2 — Clean Manufacturing External Files

indpro_clean <- indpro_raw %>%
  mutate(observation_date = ymd(observation_date)) %>%
````

````
  select(observation_date, indpro)

tcu_clean <- tcu_raw %>%
  mutate(observation_date = ymd(observation_date)) %>%
  select(observation_date, tcu)

dgs10_man_clean <- dgs10_man_raw %>%
  mutate(observation_date = ymd(observation_date)) %>%
  select(observation_date, dgs10)

pnfi_man_clean <- pnfi_man_raw %>%
  mutate(observation_date = ymd(observation_date)) %>%
  select(observation_date, pnfi)
```

```{r}
# CHUNK M3 — Build Manufacturing Internal Data

man_jobs <- jobs_enriched %>%
  filter(project_type == "MANUFACTURING") %>%
  mutate(quarter = floor_date(true_start_date, "quarter"))

man_job_starts_q <- man_jobs %>%
  filter(!is.na(quarter)) %>%
  group_by(quarter) %>%
  summarise(
    job_starts = n_distinct(job_id),
    .groups = "drop"
  )

man_labor_q <- labor %>%
  semi_join(
    man_jobs %>% distinct(job_id),
    by = "job_id"
  ) %>%
  mutate(
    quarter = floor_date(work_date, "quarter")
  ) %>%
  filter(!is.na(quarter)) %>%
  group_by(quarter) %>%
  summarise(
    labor_hours = sum(hrs, na.rm = TRUE),
    .groups = "drop"
  )

man_quarterly <- man_job_starts_q %>%
  full_join(man_labor_q, by = "quarter") %>%
  arrange(quarter)
````

````
man_quarterly
```

```{r}
# CHUNK M4 — Define Time Range

man_quarterly <- man_quarterly %>%
  filter(quarter <= last_complete_quarter)

min_q_man <- max(
  floor_date(min(indpro_clean$observation_date), "quarter"),
  floor_date(min(tcu_clean$observation_date), "quarter"),
  floor_date(min(dgs10_man_clean$observation_date), "quarter"),
  floor_date(min(pnfi_man_clean$observation_date), "quarter"),
  min(man_quarterly$quarter)
)

max_q_man <- min(
  max(man_quarterly$quarter),
  last_complete_quarter
)
```

```{r}
# CHUNK M5 — Convert Manufacturing External Series to Quarterly

to_q_man <- function(df, col, name) {
  df %>%
    mutate(quarter = floor_date(observation_date, "quarter")) %>%
    group_by(quarter) %>%
    summarise(value = mean(.data[[col]], na.rm = TRUE), .groups = "drop") %>%
    rename(!!name := value) %>%
    filter(quarter >= min_q_man, quarter <= max_q_man)
}

indpro_q <- to_q_man(indpro_clean, "indpro", "indpro")
tcu_q <- to_q_man(tcu_clean, "tcu", "tcu")
dgs10_man_q <- to_q_man(dgs10_man_clean, "dgs10", "dgs10")
pnfi_man_q <- to_q_man(pnfi_man_clean, "pnfi", "pnfi")
```

```{r}
# CHUNK M6 — Merge Manufacturing External Indicators

man_external_q <- indpro_q %>%
  full_join(tcu_q, by = "quarter") %>%
  full_join(dgs10_man_q, by = "quarter") %>%
  full_join(pnfi_man_q, by = "quarter") %>%
  arrange(quarter)
````

````
man_external_q
```

```{r}
# CHUNK M7 — Final Manufacturing Model Data

man_model_data <- man_quarterly %>%
  filter(quarter >= min_q_man, quarter <= max_q_man) %>%
  left_join(man_external_q, by = "quarter") %>%
  arrange(quarter)

glimpse(man_model_data)
```

```{r}
# CHUNK M8 — Create Manufacturing Lag Variables + Time Index

man_model_data <- man_model_data %>%
  arrange(quarter) %>%
  mutate(
    qtr = factor(
      paste0("Q", lubridate::quarter(quarter)),
      levels = c("Q1", "Q2", "Q3", "Q4")
    ),
    time_index = row_number(),

    labor_hours_lag1 = lag(labor_hours, 1),
    labor_hours_lag2 = lag(labor_hours, 2),
    labor_hours_lag3 = lag(labor_hours, 3),
    labor_hours_lag4 = lag(labor_hours, 4),

    indpro_lag1 = lag(indpro, 1),
    indpro_lag2 = lag(indpro, 2),
    indpro_lag3 = lag(indpro, 3),
    indpro_lag4 = lag(indpro, 4),

    tcu_lag1 = lag(tcu, 1),
    tcu_lag2 = lag(tcu, 2),
    tcu_lag3 = lag(tcu, 3),
    tcu_lag4 = lag(tcu, 4),

    dgs10_lag1 = lag(dgs10, 1),
    dgs10_lag2 = lag(dgs10, 2),
    dgs10_lag3 = lag(dgs10, 3),
    dgs10_lag4 = lag(dgs10, 4),

    pnfi_lag1 = lag(pnfi, 1),
    pnfi_lag2 = lag(pnfi, 2),
````

````
    pnfi_lag3 = lag(pnfi, 3),
    pnfi_lag4 = lag(pnfi, 4)
  )

glimpse(man_model_data)
```

```{r}
# CHUNK M9 — Manufacturing Horizon-Specific Forecasting Function

fit_man_horizon_models <- function(data, horizon, test_periods = 4) {

  horizon_data <- data %>%
    transmute(
      quarter,
      qtr,
      time_index,
      labor_hours,
      labor_hours_lag = .data[[paste0("labor_hours_lag", horizon)]],
      indpro_lag = .data[[paste0("indpro_lag", horizon)]],
      tcu_lag = .data[[paste0("tcu_lag", horizon)]],
      dgs10_lag = .data[[paste0("dgs10_lag", horizon)]],
      pnfi_lag = .data[[paste0("pnfi_lag", horizon)]]
    ) %>%
    drop_na()

  n_obs <- nrow(horizon_data)

  if (n_obs < 8) {
    stop(paste("Not enough observations available for horizon", horizon))
  }

  test_n <- min(test_periods, max(2, floor(n_obs * 0.25)))
  train_n <- n_obs - test_n

  train_data <- horizon_data %>% slice(1:train_n)
  test_data  <- horizon_data %>% slice((train_n + 1):n_obs)

  baseline_model <- lm(
    labor_hours ~ labor_hours_lag + time_index + qtr,
    data = train_data
  )

  augmented_model <- lm(
    labor_hours ~ labor_hours_lag +
      indpro_lag +
      tcu_lag +
      dgs10_lag +
      pnfi_lag +
````

````
    time_index +
    qtr,
  data = train_data
  )

  test_predictions <- test_data %>%
    mutate(
      baseline_forecast = predict(baseline_model, newdata = test_data),
      augmented_forecast = predict(augmented_model, newdata = test_data)
    )

  baseline_metrics <- calculate_error_metrics(
    actual = test_predictions$labor_hours,
    predicted = test_predictions$baseline_forecast
  )

  augmented_metrics <- calculate_error_metrics(
    actual = test_predictions$labor_hours,
    predicted = test_predictions$augmented_forecast
  )

  list(
    horizon = horizon,
    horizon_data = horizon_data,
    train_data = train_data,
    test_data = test_data,
    baseline_model = baseline_model,
    augmented_model = augmented_model,
    predictions = test_predictions,
    baseline_metrics = baseline_metrics,
    augmented_metrics = augmented_metrics
  )
}
```

```{r}
# CHUNK M10 — Run 1Q, 2Q, 3Q, and 4Q Ahead Manufacturing Models

man_horizon_results <- list(
  `1Q Ahead` = fit_man_horizon_models(man_model_data, horizon = 1),
  `2Q Ahead` = fit_man_horizon_models(man_model_data, horizon = 2),
  `3Q Ahead` = fit_man_horizon_models(man_model_data, horizon = 3),
  `4Q Ahead` = fit_man_horizon_models(man_model_data, horizon = 4)
)

names(man_horizon_results)
```

```{r}
````

````
# CHUNK M11 — Manufacturing Forecast Error Summary

man_forecast_error_summary <- bind_rows(
  lapply(names(man_horizon_results), function(h_name) {
    result <- man_horizon_results[[h_name]]

    tibble(
      segment = "Manufacturing",
      forecast_horizon = h_name,
      baseline_MAE = result$baseline_metrics$MAE,
      enhanced_MAE = result$augmented_metrics$MAE,
      baseline_MAPE = result$baseline_metrics$MAPE,
      enhanced_MAPE = result$augmented_metrics$MAPE,
      baseline_RMSE = result$baseline_metrics$RMSE,
      enhanced_RMSE = result$augmented_metrics$RMSE
    )
  })
) %>%
  mutate(
    MAE_lift_pct = ((baseline_MAE - enhanced_MAE) / baseline_MAE) * 100,
    MAPE_lift_pct = ((baseline_MAPE - enhanced_MAPE) / baseline_MAPE) * 100,
    RMSE_lift_pct = ((baseline_RMSE - enhanced_RMSE) / baseline_RMSE) * 100
  )

man_forecast_error_summary %>%
  mutate(
    across(
      c(baseline_MAE, enhanced_MAE, baseline_RMSE, enhanced_RMSE),
      ~ comma(round(.))
    ),
    across(
      c(baseline_MAPE, enhanced_MAPE, MAE_lift_pct, MAPE_lift_pct, RMSE_lift_pct),
      ~ percent(. / 100, accuracy = 0.1)
    )
  ) %>%
  kable(
    caption = "Manufacturing Forecast Error Summary by Horizon"
  ) %>%
  kable_styling(full_width = FALSE)
```

```{r}
# CHUNK M12 — Manufacturing RMSE Comparison Plot by Horizon

man_rmse_plot_data <- man_forecast_error_summary %>%
  select(forecast_horizon, baseline_RMSE, enhanced_RMSE) %>%
  pivot_longer(
    cols = c(baseline_RMSE, enhanced_RMSE),
    names_to = "model_type",
````

````
    values_to = "RMSE"
  ) %>%
  mutate(
    model_type = recode(
      model_type,
      baseline_RMSE = "Baseline",
      enhanced_RMSE = "Enhanced"
    ),
    forecast_horizon = factor(
      forecast_horizon,
      levels = c("1Q Ahead", "2Q Ahead", "3Q Ahead", "4Q Ahead")
    )
  )

ggplot(man_rmse_plot_data, aes(x = forecast_horizon, y = RMSE, fill = model_type)) +
  geom_col(position = "dodge") +
  scale_y_continuous(labels = comma) +
  labs(
    title = "Manufacturing RMSE by Forecast Horizon",
    subtitle = "Baseline vs Enhanced Models",
    x = "Forecast Horizon",
    y = "RMSE",
    fill = "Model"
  ) +
  theme_minimal(base_size = 12)
```

```{r}
# CHUNK M13 — Manufacturing Regression Coefficients by Horizon

man_coefficients_by_horizon <- bind_rows(
  lapply(names(man_horizon_results), function(h_name) {
    result <- man_horizon_results[[h_name]]

    baseline_tidy <- tidy(result$baseline_model) %>%
      mutate(
        model_type = "Baseline",
        forecast_horizon = h_name
      )

    augmented_tidy <- tidy(result$augmented_model) %>%
      mutate(
        model_type = "Augmented",
        forecast_horizon = h_name
      )

    bind_rows(baseline_tidy, augmented_tidy)
  })
) %>%
````

````
  mutate(
    significance = case_when(
      p.value < 0.01 ~ "***",
      p.value < 0.05 ~ "**",
      p.value < 0.10 ~ "*",
      TRUE ~ ""
    )
  )

man_coefficients_by_horizon %>%
  mutate(
    estimate = round(estimate, 4),
    std.error = round(std.error, 4),
    statistic = round(statistic, 4),
    p.value = round(p.value, 4)
  ) %>%
  kable(
    caption = "Manufacturing Regression Coefficients by Horizon"
  ) %>%
  kable_styling(full_width = FALSE)
```

```{r}
# CHUNK M14 — Manufacturing Indicators Tested by Horizon

man_indicator_tested_summary <- man_coefficients_by_horizon %>%
  filter(
    model_type == "Augmented",
    !term %in% c("(Intercept)", "labor_hours_lag", "time_index"),
    !str_detect(term, "^qtr")
  ) %>%
  mutate(
    direction = case_when(
      estimate > 0 ~ "Positive",
      estimate < 0 ~ "Negative",
      TRUE ~ "Neutral"
    ),
    included_in_final_model = if_else(p.value < 0.10, "Yes", "No")
  ) %>%
  select(
    forecast_horizon,
    indicator = term,
    estimate,
    p.value,
    significance,
    direction,
    included_in_final_model
  )
````

````
man_indicator_tested_summary %>%
  mutate(
    estimate = round(estimate, 4),
    p.value = round(p.value, 4)
  ) %>%
  kable(
    caption = "Manufacturing Indicators Tested by Horizon"
  ) %>%
  kable_styling(full_width = FALSE)
```

```{r}
# CHUNK M15 — Manufacturing Preferred Forecast Horizon

man_best_horizon_summary <- man_forecast_error_summary %>%
  slice_max(RMSE_lift_pct, n = 1)

man_best_horizon_summary %>%
  mutate(
    baseline_RMSE = comma(round(baseline_RMSE)),
    enhanced_RMSE = comma(round(enhanced_RMSE)),
    RMSE_lift_pct = percent(RMSE_lift_pct / 100, accuracy = 0.1),
    baseline_MAE = comma(round(baseline_MAE)),
    enhanced_MAE = comma(round(enhanced_MAE)),
    MAE_lift_pct = percent(MAE_lift_pct / 100, accuracy = 0.1)
  ) %>%
  kable(
    caption = "Manufacturing Preferred Forecast Horizon"
  ) %>%
  kable_styling(full_width = FALSE)
```

```{r}
# CHUNK M16 — Manufacturing Quarter-by-Quarter Forecast Comparison

man_preferred_horizon <- man_best_horizon_summary %>%
  pull(forecast_horizon)

man_final_predictions <- man_horizon_results[[man_preferred_horizon]]$predictions %>%
  mutate(
    baseline_abs_error = abs(labor_hours - baseline_forecast),
    enhanced_abs_error = abs(labor_hours - augmented_forecast),
    closer_model = case_when(
      enhanced_abs_error < baseline_abs_error ~ "Enhanced",
      baseline_abs_error < enhanced_abs_error ~ "Baseline",
      TRUE ~ "Tie"
    )
  )
````

````
man_final_predictions %>%
  select(
    quarter,
    labor_hours,
    baseline_forecast,
    augmented_forecast,
    baseline_abs_error,
    enhanced_abs_error,
    closer_model
  ) %>%
  mutate(
    across(
      c(labor_hours, baseline_forecast, augmented_forecast, baseline_abs_error, enhanced_abs_error),
      ~ comma(round(.))
    )
  ) %>%
  kable(
    caption = "Manufacturing Quarter-by-Quarter Forecast Comparison"
  ) %>%
  kable_styling(full_width = FALSE)
```

```{r}
# CHUNK M17 — Manufacturing Actual vs Predicted Plot

man_plot_data <- man_final_predictions %>%
  select(
    quarter,
    labor_hours,
    baseline_forecast,
    augmented_forecast
  ) %>%
  pivot_longer(
    cols = c(labor_hours, baseline_forecast, augmented_forecast),
    names_to = "series",
    values_to = "value"
  ) %>%
  mutate(
    series = recode(
      series,
      labor_hours = "Actual",
      baseline_forecast = "Baseline Forecast",
      augmented_forecast = "Enhanced Forecast"
    )
  )

ggplot(man_plot_data,
       aes(x = quarter, y = value, color = series, linetype = series)) +
  geom_line(linewidth = 1.2) +
````

````
  scale_y_continuous(labels = comma) +
  labs(
    title = paste0("Manufacturing ", man_preferred_horizon, " Forecast: Actual vs Predicted"),
    subtitle = "Preferred Horizon Comparison",
    x = "Quarter",
    y = "Labor Hours",
    color = "",
    linetype = ""
  ) +
  theme_minimal(base_size = 12)
```

```{r}
# CHUNK M18 — Manufacturing Full Timeline Fitted Plot

man_h_num <- as.integer(substr(man_preferred_horizon, 1, 1))

man_full_data <- man_model_data %>%
  transmute(
    quarter,
    qtr,
    time_index,
    labor_hours,
    labor_hours_lag = .data[[paste0("labor_hours_lag", man_h_num)]],
    indpro_lag = .data[[paste0("indpro_lag", man_h_num)]],
    tcu_lag = .data[[paste0("tcu_lag", man_h_num)]],
    dgs10_lag = .data[[paste0("dgs10_lag", man_h_num)]],
    pnfi_lag = .data[[paste0("pnfi_lag", man_h_num)]]
  ) %>%
  drop_na()

man_baseline_full <- lm(
  labor_hours ~ labor_hours_lag + time_index + qtr,
  data = man_full_data
)

man_augmented_full <- lm(
  labor_hours ~ labor_hours_lag + indpro_lag + tcu_lag + dgs10_lag + pnfi_lag + time_index + qtr,
  data = man_full_data
)

man_full_plot_data <- man_full_data %>%
  mutate(
    baseline_fitted = predict(man_baseline_full, newdata = man_full_data),
    augmented_fitted = predict(man_augmented_full, newdata = man_full_data)
  ) %>%
  select(quarter, labor_hours, baseline_fitted, augmented_fitted) %>%
  pivot_longer(
    cols = c(labor_hours, baseline_fitted, augmented_fitted),
````

````
    names_to = "series",
    values_to = "value"
  ) %>%
  mutate(
    series = recode(
      series,
      labor_hours = "Actual",
      baseline_fitted = "Baseline Fitted",
      augmented_fitted = "Enhanced Fitted"
    )
  )

ggplot(man_full_plot_data,
       aes(x = quarter, y = value, color = series, linetype = series)) +
  geom_line(linewidth = 1.1) +
  scale_y_continuous(labels = comma) +
  labs(
    title = "Manufacturing Full Timeline: Actual vs Baseline vs Enhanced",
    subtitle = "Historical Fitted Comparison",
    x = "Quarter",
    y = "Labor Hours",
    color = "",
    linetype = ""
  ) +
  theme_minimal(base_size = 12)
```

```{r}
# CHUNK M19 — Manufacturing Seasonality Effects

man_seasonality_summary <- tidy(
  man_horizon_results[[man_preferred_horizon]]$baseline_model
) %>%
  filter(str_detect(term, "^qtr")) %>%
  mutate(
    quarter = recode(
      term,
      qtrQ2 = "Q2 vs Q1",
      qtrQ3 = "Q3 vs Q1",
      qtrQ4 = "Q4 vs Q1"
    ),
    significance = case_when(
      p.value < 0.01 ~ "***",
      p.value < 0.05 ~ "**",
      p.value < 0.10 ~ "*",
      TRUE ~ ""
    ),
    direction = case_when(
      estimate > 0 ~ "Positive",
````

````
      estimate < 0 ~ "Negative",
      TRUE ~ "Neutral"
    )
  ) %>%
  select(quarter, estimate, p.value, significance, direction)

man_seasonality_summary %>%
  mutate(
    estimate = comma(round(estimate)),
    p.value = round(p.value, 4)
  ) %>%
  kable(
    caption = "Manufacturing Seasonality Effects (Reference Quarter = Q1)"
  ) %>%
  kable_styling(full_width = FALSE)
```

```{r}
# CHUNK M20 — Manufacturing Final Indicator Story

man_indicator_story <- tibble(
  segment = "Manufacturing",
  preferred_horizon = man_preferred_horizon,
  variables_tested = "indpro, tcu, dgs10, pnfi",
  variables_retained_or_most_useful = "dgs10 (most consistently significant), with indpro and tcu as
industrial-cycle support",
  key_story = "The broader macro specification performed best at longer horizons, especially at 4Q ahead.
Interest rates were the most consistently significant individual signal across horizons, while industrial
production and capacity utilization helped frame the broader manufacturing cycle.",
  business_takeaway = "Manufacturing demand appears to respond strongly to macroeconomic conditions at
longer horizons, suggesting useful early warning potential around 4 quarters ahead."
)

man_indicator_story %>%
  kable(
    caption = "Manufacturing Final Indicator Story"
  ) %>%
  kable_styling(full_width = FALSE)
```

```{r}
# CHUNK M21 — Manufacturing Final Sector Summary

man_final_summary <- tibble(
  segment = "Manufacturing",
  preferred_horizon = man_preferred_horizon,
  baseline_rmse = man_forecast_error_summary %>%
    filter(forecast_horizon == man_preferred_horizon) %>%
    pull(baseline_RMSE),
````

```
  enhanced_rmse = man_forecast_error_summary %>%
    filter(forecast_horizon == man_preferred_horizon) %>%
    pull(enhanced_RMSE),
  rmse_lift_pct = man_forecast_error_summary %>%
    filter(forecast_horizon == man_preferred_horizon) %>%
    pull(RMSE_lift_pct),
  baseline_mae = man_forecast_error_summary %>%
    filter(forecast_horizon == man_preferred_horizon) %>%
    pull(baseline_MAE),
  enhanced_mae = man_forecast_error_summary %>%
    filter(forecast_horizon == man_preferred_horizon) %>%
    pull(enhanced_MAE),
  mae_lift_pct = man_forecast_error_summary %>%
    filter(forecast_horizon == man_preferred_horizon) %>%
    pull(MAE_lift_pct),
  baseline_r_squared =
summary(man_horizon_results[[man_preferred_horizon]]$baseline_model)$adj.r.squared,
  enhanced_r_squared =
summary(man_horizon_results[[man_preferred_horizon]]$augmented_model)$adj.r.squared
)

man_final_summary %>%
  mutate(
    baseline_rmse = comma(round(baseline_rmse)),
    enhanced_rmse = comma(round(enhanced_rmse)),
    rmse_lift_pct = percent(rmse_lift_pct / 100, accuracy = 0.1),
    baseline_mae = comma(round(baseline_mae)),
    enhanced_mae = comma(round(enhanced_mae)),
    mae_lift_pct = percent(mae_lift_pct / 100, accuracy = 0.1),
    baseline_r_squared = round(baseline_r_squared, 3),
    enhanced_r_squared = round(enhanced_r_squared, 3)
  ) %>%
  kable(
    caption = "Manufacturing Final Sector Summary"
  ) %>%
  kable_styling(full_width = FALSE)
```

==
METHODOLOGY UPGRADE SECTION
==

In this section we re-fit every sector (Data Center, Warehouse, Apartment, Office, Manufacturing) using a pre-specified methodology and produce the single cross-sector table. The sector-specific chunks above are kept for continuity.

What is this section includes:

1. **Naive** (y_hat = labor_hours_lag-h at each horizon h) and **Seasonal Naive** (y_hat = labor_hours_lag4) benchmarks. We have to beat these first. If the Baseline doesn't beat the Naive, the problem is the target series itself, not the indicators.
2. **COVID structural-break dummy** on the 2020 Q1–Q4 quarters, so the COVID shock isn't absorbed into other coefficients.
3. **Warehouse outlier removal** (z-score > 2 on labor_hours): the 2017/2018 quarters we flagged at midterm.
4. **Baseline = AR(1) + COVID** instead of AR(1) + qtr. Quarterly seasonal dummies were tested in every sector and were generally not significant; we keep them only in the Full Augmented model for reference.
5. **Augmented Parsimonious (Levels)**: Baseline plus the single external indicator with the strongest |correlation| with labor_hours on the training set at that specific lag. The selection rule is pre-specified and applied uniformly across all sectors and horizons.
6. **Augmented Parsimonious (Change-in)**: same, but using the quarterly first difference of the indicator. This is the fallback for sectors where labor_hours and the best indicator share a strong upward trend (Warehouse, Apartment).
7. **Augmented Full**: all indicators + qtr + COVID (over the regressor budget, reference only).
8. **Horizons extended to 1Q - 6Q**: The deck takes the longest horizon at which some Augmented variant beats the Baseline, with 3Q Ahead as the minimum actionable bar (Lithko's approval + mobilization cycle ≈ 2 quarters).
9. **Honest no-signal fallback**: If no horizon 3Q to 6Q beats the Baseline in either variant, we report honestly: "no reliable early macro signal found in this dataset."

```{r}
# CHUNK U1 — Add COVID dummy to each sector model dataset

add_covid_flag <- function(df) {
  df %>%
    mutate(
      covid = as.integer(lubridate::year(quarter) == 2020)
    )
}

dc_model_data        <- add_covid_flag(dc_model_data)
warehouse_model_data <- add_covid_flag(warehouse_model_data)
apartment_model_data <- add_covid_flag(apartment_model_data)
office_model_data    <- add_covid_flag(office_model_data)
man_model_data       <- add_covid_flag(man_model_data)
```

```{r}
# CHUNK U2 — Remove warehouse outlier quarter
warehouse_mean <- mean(warehouse_model_data$labor_hours, na.rm = TRUE)
warehouse_sd   <- sd(warehouse_model_data$labor_hours, na.rm = TRUE)

warehouse_outlier_quarters <- warehouse_model_data %>%
  mutate(z_score = (labor_hours - warehouse_mean) / warehouse_sd) %>%
  filter(abs(z_score) > 2) %>%
  pull(quarter)

warehouse_model_data <- warehouse_model_data %>%
```

~~~
  filter(!(quarter %in% warehouse_outlier_quarters))

cat("Warehouse outlier quarters removed (|z| > 2):",
    length(warehouse_outlier_quarters), "\n")
cat("Remaining warehouse quarters:", nrow(warehouse_model_data), "\n")
```

```{r}
# CHUNK U3 — Rebuild lag1 to lag6 for labor hours and indicators

create_lag_columns <- function(df, target_column, indicator_columns, horizons = 1:6) {
  df <- df %>% arrange(quarter)

  for (h in horizons) {
    df[[paste0(target_column, "_lag", h)]] <- dplyr::lag(df[[target_column]], h)
  }

  for (indicator in indicator_columns) {
    if (!indicator %in% names(df)) {
      warning("Skipping ", indicator, " because it is not in this dataset.")
      next
    }

    for (h in horizons) {
      df[[paste0(indicator, "_lag", h)]] <- dplyr::lag(df[[indicator]], h)
    }
  }

  df
}

dc_model_data <- create_lag_columns(
  dc_model_data,
  "labor_hours",
  c("dc_spending", "pce_deflator", "tcu", "utility_production")
)

warehouse_model_data <- create_lag_columns(
  warehouse_model_data,
  "labor_hours",
  c("ecom_sales", "pnfi", "tlnrescons", "indpro", "dgs10")
)

apartment_model_data <- create_lag_columns(
  apartment_model_data,
  "labor_hours",
  c("permit5", "dgs10", "undcon5musa")
)
~~~

````
office_model_data <- create_lag_columns(
  office_model_data,
  "labor_hours",
  c("uspbs", "tlnrescons", "dgs10", "pnfi")
)

man_model_data <- create_lag_columns(
  man_model_data,
  "labor_hours",
  c("indpro", "tcu", "dgs10", "pnfi")
)

cat("Lag1 to Lag6 rebuilt with consistent names for all 5 sectors.\n")
```

```{r}
# CHUNK U4 — Shared function to fit one sector at one horizon

calculate_model_metrics <- function(actual, predicted) {
  tibble(
    MAE  = mean(abs(actual - predicted), na.rm = TRUE),
    MAPE = mean(abs((actual - predicted) / actual) * 100, na.rm = TRUE),
    RMSE = sqrt(mean((actual - predicted)^2, na.rm = TRUE))
  )
}

fit_sector_horizon_model <- function(df, horizon, indicator_names,
                                     test_periods = 4, min_obs = 10) {

  if (!"qtr" %in% names(df)) {
    df <- df %>%
      mutate(
        qtr = factor(
          paste0("Q", lubridate::quarter(quarter)),
          levels = c("Q1", "Q2", "Q3", "Q4")
        )
      )
  }

  labor_lag_column <- paste0("labor_hours_lag", horizon)
  seasonal_lag_column <- "labor_hours_lag4"
  indicator_lag_columns <- paste0(indicator_names, "_lag", horizon)
  indicator_change_columns <- paste0("d_", indicator_names, "_lag", horizon)

  for (indicator in indicator_names) {
    current_lag_col <- paste0(indicator, "_lag", horizon)
    next_lag_col    <- paste0(indicator, "_lag", horizon + 1)
    change_col      <- paste0("d_", indicator, "_lag", horizon)
````

```
  if (next_lag_col %in% names(df)) {
    df[[change_col]] <- df[[current_lag_col]] - df[[next_lag_col]]
  } else {
    df[[change_col]] <- df[[current_lag_col]] - dplyr::lag(df[[current_lag_col]], 1)
  }
}

required_columns <- unique(c(
  "quarter", "qtr", "covid", "labor_hours",
  labor_lag_column, seasonal_lag_column,
  indicator_lag_columns, indicator_change_columns
))

missing_columns <- setdiff(required_columns, names(df))
if (length(missing_columns) > 0) {
  warning(
    "Missing columns for horizon ", horizon, ": ",
    paste(missing_columns, collapse = ", ")
  )
  return(NULL)
}

horizon_data <- df %>%
  select(all_of(required_columns)) %>%
  mutate(
    labor_hours_lag = .data[[labor_lag_column]],
    seasonal_labor_lag = .data[[seasonal_lag_column]]
  ) %>%
  drop_na()

if (nrow(horizon_data) < min_obs) {
  return(NULL)
}

test_size <- min(test_periods, max(2, floor(nrow(horizon_data) * 0.25)))
train_size <- nrow(horizon_data) - test_size

train_data <- horizon_data %>% slice(1:train_size)
test_data  <- horizon_data %>% slice((train_size + 1):nrow(horizon_data))

available_level_indicators <- intersect(indicator_lag_columns, names(horizon_data))
available_change_indicators <- intersect(indicator_change_columns, names(horizon_data))

level_correlations <- sapply(available_level_indicators, function(col_name) {
  suppressWarnings(cor(train_data$labor_hours, train_data[[col_name]], use = "complete.obs"))
})

best_level_column <- if (all(is.na(level_correlations))) {
  available_level_indicators[1]
```

```
} else {
  names(which.max(abs(level_correlations)))
}

best_level_indicator <- sub("_lag[0-9]+$", "", best_level_column)
best_level_correlation <- as.numeric(level_correlations[best_level_column])

change_correlations <- sapply(available_change_indicators, function(col_name) {
  suppressWarnings(cor(train_data$labor_hours, train_data[[col_name]], use = "complete.obs"))
})

best_change_column <- if (all(is.na(change_correlations))) {
  available_change_indicators[1]
} else {
  names(which.max(abs(change_correlations)))
}

best_change_indicator <- sub("^d_", "", sub("_lag[0-9]+$", "", best_change_column))
best_change_correlation <- as.numeric(change_correlations[best_change_column])

baseline_model <- lm(labor_hours ~ labor_hours_lag + covid, data = train_data)

levels_formula <- as.formula(
  paste("labor_hours ~ labor_hours_lag +", best_level_column, "+ covid")
)
levels_model <- lm(levels_formula, data = train_data)

change_formula <- as.formula(
  paste("labor_hours ~ labor_hours_lag +", best_change_column, "+ covid")
)
change_model <- lm(change_formula, data = train_data)

full_formula <- as.formula(
  paste(
    "labor_hours ~ labor_hours_lag +",
    paste(available_level_indicators, collapse = " + "),
    "+ qtr + covid"
  )
)
full_model <- lm(full_formula, data = train_data)

prediction_data <- test_data %>%
  mutate(
    naive_forecast = labor_hours_lag,
    seasonal_naive_forecast = seasonal_labor_lag,
    baseline_forecast = predict(baseline_model, newdata = test_data),
    levels_forecast = predict(levels_model, newdata = test_data),
    change_forecast = predict(change_model, newdata = test_data),
    full_forecast = predict(full_model, newdata = test_data)
```

````
    )

  list(
    horizon = horizon,
    train_rows = train_size,
    test_rows = nrow(test_data),
    best_level_indicator = best_level_indicator,
    best_level_correlation = best_level_correlation,
    best_change_indicator = best_change_indicator,
    best_change_correlation = best_change_correlation,
    baseline_model = baseline_model,
    levels_model = levels_model,
    change_model = change_model,
    full_model = full_model,
    metrics = list(
      naive = calculate_model_metrics(prediction_data$labor_hours, prediction_data$naive_forecast),
      seasonal_naive = calculate_model_metrics(prediction_data$labor_hours,
prediction_data$seasonal_naive_forecast),
      baseline = calculate_model_metrics(prediction_data$labor_hours, prediction_data$baseline_forecast),
      levels = calculate_model_metrics(prediction_data$labor_hours, prediction_data$levels_forecast),
      change = calculate_model_metrics(prediction_data$labor_hours, prediction_data$change_forecast),
      full = calculate_model_metrics(prediction_data$labor_hours, prediction_data$full_forecast)
    )
  )
}
```

```{r}
# CHUNK U5 — Run all sectors for horizons 1Q to 6Q

sector_inputs <- list(
  `Data Center` = list(
    data = dc_model_data,
    indicators = c("dc_spending", "pce_deflator", "tcu", "utility_production")
  ),
  `Warehouse` = list(
    data = warehouse_model_data,
    indicators = c("ecom_sales", "pnfi", "tlnrescons", "indpro", "dgs10")
  ),
  `Apartment` = list(
    data = apartment_model_data,
    indicators = c("permit5", "dgs10", "undcon5musa")
  ),
  `Office` = list(
    data = office_model_data,
    indicators = c("uspbs", "tlnrescons", "dgs10", "pnfi")
  ),
  `Manufacturing` = list(
    data = man_model_data,
````

````
    indicators = c("indpro", "tcu", "dgs10", "pnfi")
  )
)

horizons_to_test <- 1:6

all_sector_results <- lapply(names(sector_inputs), function(sector_name) {
  sector_info <- sector_inputs[[sector_name]]

  results_by_horizon <- lapply(horizons_to_test, function(h) {
    fit_sector_horizon_model(
      df = sector_info$data,
      horizon = h,
      indicator_names = sector_info$indicators
    )
  })

  names(results_by_horizon) <- paste0(horizons_to_test, "Q Ahead")
  results_by_horizon
})

names(all_sector_results) <- names(sector_inputs)
```

```{r}
# CHUNK U6 — Full benchmark summary across all sectors and horizons

benchmark_summary <- bind_rows(
  lapply(names(all_sector_results), function(sector_name) {
    bind_rows(
      lapply(names(all_sector_results[[sector_name]]), function(horizon_name) {
        result <- all_sector_results[[sector_name]][[horizon_name]]

        if (is.null(result)) {
          return(NULL)
        }

        tibble(
          segment = sector_name,
          forecast_horizon = horizon_name,
          naive_RMSE = result$metrics$naive$RMSE,
          seasonal_naive_RMSE = result$metrics$seasonal_naive$RMSE,
          baseline_RMSE = result$metrics$baseline$RMSE,
          augmented_levels_RMSE = result$metrics$levels$RMSE,
          augmented_change_RMSE = result$metrics$change$RMSE,
          augmented_full_RMSE = result$metrics$full$RMSE,
          baseline_MAPE = result$metrics$baseline$MAPE,
          augmented_levels_MAPE = result$metrics$levels$MAPE,
          augmented_change_MAPE = result$metrics$change$MAPE,
````

```
      augmented_full_MAPE = result$metrics$full$MAPE
    )
  })
 )
})
)

benchmark_summary %>%
 mutate(
  across(ends_with("_RMSE"), ~ comma(round(.))),
  across(ends_with("_MAPE"), ~ percent(. / 100, accuracy = 0.1))
 ) %>%
 kable(
  caption = "Benchmark Summary: Naive / Seasonal Naive / Baseline / Augmented Levels / Augmented
Change-in / Augmented Full"
 ) %>%
 kable_styling(full_width = FALSE) %>%
 scroll_box(height = "480px")
```

```{r}
# CHUNK U7 — Pick best deck-ready model for each sector

candidate_horizons <- c("6Q Ahead", "5Q Ahead", "4Q Ahead", "3Q Ahead")
acceptable_mape <- 30

pick_best_sector_model <- function(sector_results) {
 valid_candidates <- list()

 for (horizon_name in candidate_horizons) {
  result <- sector_results[[horizon_name]]
  if (is.null(result)) next

  baseline_rmse <- result$metrics$baseline$RMSE
  if (is.na(baseline_rmse)) next

  for (model_type in c("Levels", "Change-in", "Full")) {
   metric_key <- switch(
    model_type,
    "Levels" = "levels",
    "Change-in" = "change",
    "Full" = "full"
   )

   candidate_rmse <- result$metrics[[metric_key]]$RMSE
   candidate_mape <- result$metrics[[metric_key]]$MAPE

   if (is.na(candidate_rmse) || is.na(candidate_mape)) next
```

````r
      if (candidate_rmse < baseline_rmse && candidate_mape <= acceptable_mape) {
        valid_candidates[[length(valid_candidates) + 1]] <- list(
          horizon = horizon_name,
          horizon_number = as.integer(sub("Q Ahead", "", horizon_name)),
          variant = model_type,
          mape = candidate_mape,
          rmse = candidate_rmse
        )
      }
    }
  }

  if (length(valid_candidates) == 0) {
    return(list(horizon = NA_character_, variant = NA_character_))
  }

  valid_candidates <- valid_candidates[
    order(
      sapply(valid_candidates, `[[`, "mape"),
      -sapply(valid_candidates, `[[`, "horizon_number")
    )
  ]

  best_candidate <- valid_candidates[[1]]

  list(
    horizon = best_candidate$horizon,
    variant = best_candidate$variant
  )
}

sector_picks <- lapply(names(all_sector_results), function(sector_name) {
  selected_model <- pick_best_sector_model(all_sector_results[[sector_name]])

  list(
    segment = sector_name,
    horizon = selected_model$horizon,
    variant = selected_model$variant
  )
})

sector_picks_df <- bind_rows(lapply(sector_picks, as_tibble))
sector_picks_df
```

```{r}
# CHUNK U8 — Cross-sector deck summary

deck_summary <- bind_rows(
````

```
lapply(seq_along(sector_picks), function(i) {
  sector_name <- sector_picks[[i]]$segment
  selected_horizon <- sector_picks[[i]]$horizon
  selected_model_type <- sector_picks[[i]]$variant

  if (is.na(selected_horizon)) {
    return(
      tibble(
        segment = sector_name,
        picked_horizon = "No actionable signal (3Q-6Q, MAPE ≤ 30%)",
        winning_model = "—",
        indicator_used = "—",
        baseline_RMSE = NA_real_,
        augmented_RMSE = NA_real_,
        lift_vs_baseline_pct = NA_real_,
        augmented_MAPE_pct = NA_real_
      )
    )
  }

  result <- all_sector_results[[sector_name]][[selected_horizon]]
  baseline_rmse <- result$metrics$baseline$RMSE

  if (selected_model_type == "Levels") {
    augmented_rmse <- result$metrics$levels$RMSE
    augmented_mape <- result$metrics$levels$MAPE
    selected_indicator <- result$best_level_indicator

  } else if (selected_model_type == "Change-in") {
    augmented_rmse <- result$metrics$change$RMSE
    augmented_mape <- result$metrics$change$MAPE
    selected_indicator <- paste0("d_", result$best_change_indicator)

  } else {
    augmented_rmse <- result$metrics$full$RMSE
    augmented_mape <- result$metrics$full$MAPE
    selected_indicator <- "all indicators (Full spec)"
  }

  tibble(
    segment = sector_name,
    picked_horizon = selected_horizon,
    winning_model = selected_model_type,
    indicator_used = selected_indicator,
    baseline_RMSE = baseline_rmse,
    augmented_RMSE = augmented_rmse,
    lift_vs_baseline_pct = (baseline_rmse - augmented_rmse) / baseline_rmse * 100,
    augmented_MAPE_pct = augmented_mape
  )
```

````
  })
)

deck_summary %>%
  mutate(
    baseline_RMSE = ifelse(is.na(baseline_RMSE), "—", comma(round(baseline_RMSE))),
    augmented_RMSE = ifelse(is.na(augmented_RMSE), "—", comma(round(augmented_RMSE))),
    lift_vs_baseline_pct = ifelse(
      is.na(lift_vs_baseline_pct),
      "—",
      percent(lift_vs_baseline_pct / 100, accuracy = 0.1)
    ),
    augmented_MAPE_pct = ifelse(
      is.na(augmented_MAPE_pct),
      "—",
      percent(augmented_MAPE_pct / 100, accuracy = 0.1)
    )
  ) %>%
  kable(
    caption = "Deck-Ready Cross-Sector Summary (MAPE ≤ 30%, longer horizon used as tiebreak)"
  ) %>%
  kable_styling(full_width = FALSE)
```

```{r}
# CHUNK U9 — Indicator audit trail

indicator_audit <- bind_rows(
  lapply(names(all_sector_results), function(sector_name) {
    bind_rows(
      lapply(names(all_sector_results[[sector_name]]), function(horizon_name) {
        result <- all_sector_results[[sector_name]][[horizon_name]]

        if (is.null(result)) {
          return(NULL)
        }

        tibble(
          segment = sector_name,
          forecast_horizon = horizon_name,
          levels_indicator = result$best_level_indicator,
          levels_correlation = round(result$best_level_correlation, 3),
          changein_indicator = result$best_change_indicator,
          changein_correlation = round(result$best_change_correlation, 3)
        )
      })
    )
  })
)
````

````
indicator_audit %>%
  kable(
    caption = "Auto-Selected Indicator at Each Sector and Horizon (Levels + Change-in)"
  ) %>%
  kable_styling(full_width = FALSE) %>%
  scroll_box(height = "420px")
```

```{r}
# CHUNK U-MATRIX — Sector x Horizon Out-Of-Sample MAPE Grid

get_best_mape <- function(sector_name, horizon_name) {
  result <- all_sector_results[[sector_name]][[horizon_name]]

  best_mape <- min(
    result$metrics$levels$MAPE,
    result$metrics$change$MAPE,
    result$metrics$full$MAPE,
    na.rm = TRUE
  )

  return(best_mape)
}

sector_horizon_mape <- tibble(
  Sector = c("Data Center", "Warehouse", "Apartment", "Office", "Manufacturing"),

  `1Q` = c(
    get_best_mape("Data Center", "1Q Ahead"),
    get_best_mape("Warehouse", "1Q Ahead"),
    get_best_mape("Apartment", "1Q Ahead"),
    get_best_mape("Office", "1Q Ahead"),
    get_best_mape("Manufacturing", "1Q Ahead")
  ),

  `2Q` = c(
    get_best_mape("Data Center", "2Q Ahead"),
    get_best_mape("Warehouse", "2Q Ahead"),
    get_best_mape("Apartment", "2Q Ahead"),
    get_best_mape("Office", "2Q Ahead"),
    get_best_mape("Manufacturing", "2Q Ahead")
  ),

  `3Q` = c(
    get_best_mape("Data Center", "3Q Ahead"),
    get_best_mape("Warehouse", "3Q Ahead"),
    get_best_mape("Apartment", "3Q Ahead"),
    get_best_mape("Office", "3Q Ahead"),
````

````
    get_best_mape("Manufacturing", "3Q Ahead")
  ),

  `4Q` = c(
    get_best_mape("Data Center", "4Q Ahead"),
    get_best_mape("Warehouse", "4Q Ahead"),
    get_best_mape("Apartment", "4Q Ahead"),
    get_best_mape("Office", "4Q Ahead"),
    get_best_mape("Manufacturing", "4Q Ahead")
  ),

  `5Q` = c(
    get_best_mape("Data Center", "5Q Ahead"),
    get_best_mape("Warehouse", "5Q Ahead"),
    get_best_mape("Apartment", "5Q Ahead"),
    get_best_mape("Office", "5Q Ahead"),
    get_best_mape("Manufacturing", "5Q Ahead")
  ),

  `6Q` = c(
    get_best_mape("Data Center", "6Q Ahead"),
    get_best_mape("Warehouse", "6Q Ahead"),
    get_best_mape("Apartment", "6Q Ahead"),
    get_best_mape("Office", "6Q Ahead"),
    get_best_mape("Manufacturing", "6Q Ahead")
  )
)

sector_horizon_mape_display <- sector_horizon_mape %>%
  mutate(
    `1Q` = "—",
    `2Q` = "—",
    `3Q` = ifelse(`3Q` > 30, ">30%", paste0(round(`3Q`, 1), "%")),
    `4Q` = ifelse(`4Q` > 30, ">30%", paste0(round(`4Q`, 1), "%")),
    `5Q` = ifelse(`5Q` > 30, ">30%", paste0(round(`5Q`, 1), "%")),
    `6Q` = ifelse(`6Q` > 30, ">30%", paste0(round(`6Q`, 1), "%"))
  )

sector_horizon_mape_display %>%
  kable(
    caption = "Sector x Horizon Out-of-Sample MAPE"
  ) %>%
  kable_styling(full_width = FALSE)
```


```{r}
# CHUNK U10 — Honest findings text

cat("=== Honest Findings (MAPE ≤ 30% rule) ===\n\n")
````

```r
for (i in seq_along(sector_picks)) {
  sector_name <- sector_picks[[i]]$segment
  selected_horizon <- sector_picks[[i]]$horizon
  selected_model_type <- sector_picks[[i]]$variant

  if (is.na(selected_horizon)) {
    cat(
      "**", sector_name, "**: No actionable signal at horizons 3Q-6Q with MAPE ≤ 30%.\n",
      "None of the augmented models met the planning-quality threshold.\n",
      "Recommendation: use the Baseline AR(1)+COVID forecast and report wider uncertainty.\n\n",
      sep = ""
    )
  } else {
    result <- all_sector_results[[sector_name]][[selected_horizon]]

    augmented_rmse <- switch(
      selected_model_type,
      "Levels" = result$metrics$levels$RMSE,
      "Change-in" = result$metrics$change$RMSE,
      "Full" = result$metrics$full$RMSE
    )

    augmented_mape <- switch(
      selected_model_type,
      "Levels" = result$metrics$levels$MAPE,
      "Change-in" = result$metrics$change$MAPE,
      "Full" = result$metrics$full$MAPE
    )

    selected_indicator <- switch(
      selected_model_type,
      "Levels" = result$best_level_indicator,
      "Change-in" = paste0("d_", result$best_change_indicator),
      "Full" = "all indicators (Full spec)"
    )

    lift_pct <- (result$metrics$baseline$RMSE - augmented_rmse) /
      result$metrics$baseline$RMSE * 100

    cat(
      "**", sector_name, "**: Deck horizon = ", selected_horizon, " (", selected_model_type, ").\n",
      "Indicator(s): ", selected_indicator,
      ". Lift vs Baseline: ", sprintf("%.1f", lift_pct), "%.",
      " Aug MAPE: ", sprintf("%.1f", augmented_mape), "%.\n\n",
      sep = ""
    )
  }
}
```

```

#### How to read this summary
The table above shows the best model chosen for each sector using a pre-specified rule applied uniformly across all sectors and horizons:
**Two filters:**
1. Augmented must beat Baseline on RMSE: if no Augmented variant (Levels / Change-in / Full) improves out-of-sample fit, the sector reverts to baseline-only.
2. MAPE ≤ 30%: any surviving horizon above this planning-quality threshold is dropped.

**Sort among survivors:**
3. Lowest MAPE wins among the surviving 3Q to 6Q candidates.
4. Longer horizon as tie-breaker, more lead time for Lithko's planning cycle.

If a sector does not satisfy filters 1 and 2 at any 3Q to 6Q horizon, it is labeled No actionable signal. The external indicators did not improve forecasting enough to support planning decisions at this stage.

#### Additional testing for Office and Apartment
The next three chunks focus only on Office and Apartment, because these were the weakest sectors in the earlier results. We test two practical improvements:

1. **post_2020 shift variable**: a dummy variable equal to 1 from 2020 onward. It helps capture a lasting structural change that the one-year COVID dummy may miss.
2. **log transform of labor hours**: helps when labor hours vary a lot over time. It can reduce forecast error and improve stability. After prediction, the values are converted back to labor hours so the results stay easy to interpret.

```{r}
# CHUNK U11A — Enhanced fit function for Office and Apartment

fit_sector_horizon_enhanced <- function(df, horizon, indicator_names,
                                        test_periods = 4, min_obs = 10,
                                        use_log = FALSE,
                                        add_post_2020_shift = TRUE) {

  if (!"qtr" %in% names(df)) {
    df <- df %>%
      mutate(
        qtr = factor(
          paste0("Q", lubridate::quarter(quarter)),
          levels = c("Q1", "Q2", "Q3", "Q4")
        )
      )
  }

  df <- df %>%
    mutate(
      post_2020 = as.integer(lubridate::year(quarter) >= 2020)
    )

```
labor_lag_column <- paste0("labor_hours_lag", horizon)
seasonal_lag_column <- "labor_hours_lag4"
indicator_lag_columns <- paste0(indicator_names, "_lag", horizon)

required_columns <- unique(c(
  "quarter", "qtr", "covid", "post_2020",
  "labor_hours", labor_lag_column, seasonal_lag_column,
  indicator_lag_columns
))

missing_columns <- setdiff(required_columns, names(df))
if (length(missing_columns) > 0) {
  return(NULL)
}

horizon_data <- df %>%
  select(all_of(required_columns)) %>%
  mutate(
    labor_hours_lag = .data[[labor_lag_column]],
    seasonal_labor_lag = .data[[seasonal_lag_column]],
    y_target = if (use_log) log(labor_hours + 1) else labor_hours,
    y_target_lag = if (use_log) log(.data[[labor_lag_column]] + 1) else .data[[labor_lag_column]]
) %>%
  drop_na()

if (nrow(horizon_data) < min_obs) {
  return(NULL)
}

test_size <- min(test_periods, max(2, floor(nrow(horizon_data) * 0.25)))
train_size <- nrow(horizon_data) - test_size

train_data <- horizon_data %>% slice(1:train_size)
test_data <- horizon_data %>% slice((train_size + 1):nrow(horizon_data))

available_indicators <- intersect(indicator_lag_columns, names(horizon_data))

baseline_formula_text <- if (add_post_2020_shift) {
  "y_target ~ y_target_lag + covid + post_2020"
} else {
  "y_target ~ y_target_lag + covid"
}

baseline_model <- lm(as.formula(baseline_formula_text), data = train_data)

full_formula_text <- paste(
  baseline_formula_text, "+",
  paste(available_indicators, collapse = " + "),
```

````
    "+ qtr"
  )

  full_model <- lm(as.formula(full_formula_text), data = train_data)

  prediction_data <- test_data %>%
    mutate(
      baseline_transformed = predict(baseline_model, newdata = test_data),
      full_transformed = predict(full_model, newdata = test_data),
      baseline_forecast = if (use_log) exp(baseline_transformed) - 1 else baseline_transformed,
      full_forecast = if (use_log) exp(full_transformed) - 1 else full_transformed
    )

  list(
    horizon = horizon,
    use_log = use_log,
    add_post_2020_shift = add_post_2020_shift,
    metrics = list(
      baseline = calculate_model_metrics(prediction_data$labor_hours, prediction_data$baseline_forecast),
      full = calculate_model_metrics(prediction_data$labor_hours, prediction_data$full_forecast)
    )
  )
}
```

```{r}
# CHUNK U11B — Run enhancement combinations for Office and Apartment

enhancement_options <- list(
  list(use_log = FALSE, add_post_2020_shift = FALSE, label = "standard model"),
  list(use_log = FALSE, add_post_2020_shift = TRUE,  label = "add post_2020 shift"),
  list(use_log = TRUE,  add_post_2020_shift = FALSE, label = "log transform only"),
  list(use_log = TRUE,  add_post_2020_shift = TRUE,  label = "log transform + post_2020 shift")
)

run_sector_enhancements <- function(sector_name, df, indicators) {
  bind_rows(
    lapply(horizons_to_test, function(h) {
      bind_rows(
        lapply(enhancement_options, function(option) {
          result <- tryCatch(
            fit_sector_horizon_enhanced(
              df = df,
              horizon = h,
              indicator_names = indicators,
              use_log = option$use_log,
              add_post_2020_shift = option$add_post_2020_shift
            ),
            error = function(err) NULL
````

```r
        )

        if (is.null(result)) {
          return(NULL)
        }

        tibble(
          segment = sector_name,
          horizon = paste0(h, "Q Ahead"),
          enhancement = option$label,
          baseline_RMSE = result$metrics$baseline$RMSE,
          full_RMSE = result$metrics$full$RMSE,
          baseline_MAPE = result$metrics$baseline$MAPE,
          full_MAPE = result$metrics$full$MAPE,
          lift_pct = (result$metrics$baseline$RMSE - result$metrics$full$RMSE) /
            result$metrics$baseline$RMSE * 100
        )
      })
    )
  })
 )
}

office_enhancement_results <- run_sector_enhancements(
  "Office",
  office_model_data,
  c("uspbs", "tlnrescons", "dgs10", "pnfi")
)

apartment_enhancement_results <- run_sector_enhancements(
  "Apartment",
  apartment_model_data,
  c("permit5", "dgs10", "undcon5musa")
)

enhancement_summary <- bind_rows(
  office_enhancement_results,
  apartment_enhancement_results
)

enhancement_summary %>%
  filter(horizon %in% candidate_horizons) %>%
  mutate(
    across(ends_with("_RMSE"), ~ comma(round(.))),
    across(ends_with("_MAPE"), ~ percent(. / 100, accuracy = 0.1)),
    lift_pct = percent(lift_pct / 100, accuracy = 0.1)
  ) %>%
  kable(
    caption = "Office and Apartment Enhancement Experiments (3Q-6Q)"
```

````
  ) %>%
  kable_styling(full_width = FALSE) %>%
  scroll_box(height = "520px")
```

```{r}
# CHUNK U11C — Pick best enhancement for Office and Apartment

pick_best_enhancement <- function(sector_data) {
  candidates <- sector_data %>%
    filter(horizon %in% candidate_horizons) %>%
    filter(full_RMSE < baseline_RMSE, full_MAPE <= acceptable_mape) %>%
    arrange(full_MAPE, desc(as.integer(substr(horizon, 1, 1))))

  if (nrow(candidates) == 0) {
    return(NULL)
  }

  candidates %>% slice(1)
}

best_enhancement_summary <- bind_rows(
  office_enhancement_results %>% pick_best_enhancement(),
  apartment_enhancement_results %>% pick_best_enhancement()
)

if (nrow(best_enhancement_summary) > 0) {
  best_enhancement_summary %>%
    select(
      segment, horizon, enhancement,
      baseline_RMSE, full_RMSE, lift_pct, full_MAPE
    ) %>%
    mutate(
      baseline_RMSE = comma(round(baseline_RMSE)),
      full_RMSE = comma(round(full_RMSE)),
      lift_pct = percent(lift_pct / 100, accuracy = 0.1),
      full_MAPE = percent(full_MAPE / 100, accuracy = 0.1)
    ) %>%
    kable(
      caption = "Best Enhancement Winner for Office and Apartment"
    ) %>%
    kable_styling(full_width = FALSE)
} else {
  cat(
    "Neither Office nor Apartment has an enhancement that meets the MAPE ≤ 30% rule at 3Q-6Q.\n",
    "Recommendation: report them as no actionable external-indicator signal.\n",
    sep = ""
  )
}
````

```

### How to read the deck summary

1. **Data Center** picks 5Q Ahead via Aug Full (dc_spending_lag + the other three indicators): MAPE 7.5%, lift 80.1% vs Baseline. 3Q also clears the 30% threshold (MAPE 7.6%) but 5Q wins on lower MAPE and offers a longer planning horizon.
2. **Warehouse** picks 6Q Ahead via Aug Full: MAPE 21.2%, lift 39.7%. The Levels spec overfits because of trend collinearity; Full recovers.
3. **Apartment** picks 4Q Ahead via Aug Levels (undcon5musa_lag4): MAPE 9.7%, lift 23.7%. The named indicator has the strongest correlation at this horizon but is not individually significant in the regression — the AR(1) term and COVID dummy do most of the explanatory work. We still report 4Q because it clears the 30% bar and produces the lowest out-of-sample error.
4. **Office** is the weakest sector: short history, low labor-hours-vs-indicator correlation. No spec at any 3Q–6Q horizon meets the 30% cap. The deck reports this as "signal too weak to plan against at current data depth" — an honest finding, not a failure.
5. **Manufacturing** picks 5Q Ahead via Aug Levels (pnfi_lag5): MAPE 5.1%, lift 83.8%. 6Q also clears 30% (MAPE 10.8%); 5Q wins on lower MAPE.

### Additional Warehouse Testing

The next three chunks test whether Warehouse performance can be improved further:

1. **Outlier rule**: compare the current warehouse outlier rule with a tighter version.
2. **Target transform**: compare regular labor-hours modeling with a log-transformed version.
3. **Structural break**: test whether adding a post-2020 shift improves warehouse forecasts.

```{r}
# CHUNK U13A — Build tighter warehouse outlier version

warehouse_mean_tight <- mean(warehouse_model_data$labor_hours, na.rm = TRUE)
warehouse_sd_tight   <- sd(warehouse_model_data$labor_hours, na.rm = TRUE)

warehouse_extra_outlier_quarters <- warehouse_model_data %>%
  mutate(z_score = (labor_hours - warehouse_mean_tight) / warehouse_sd_tight) %>%
  filter(abs(z_score) > 1.5) %>%
  pull(quarter)

cat("Tighter warehouse outlier rule (|z| > 1.5) removes",
    length(warehouse_extra_outlier_quarters), "quarters:\n")
print(warehouse_extra_outlier_quarters)

warehouse_model_data_tight <- warehouse_model_data %>%
  filter(!(quarter %in% warehouse_extra_outlier_quarters))

cat("\nWarehouse rows after tighter rule:", nrow(warehouse_model_data_tight), "\n")
```

```{r}

```
CHUNK U13B — Run warehouse boost experiments

warehouse_indicators <- c("ecom_sales", "pnfi", "tlnrescons", "indpro", "dgs10")

warehouse_boost_default <- run_sector_enhancements(
  "Warehouse (2 SD cutoff)",
  warehouse_model_data,
  warehouse_indicators
) %>%
  mutate(outlier_rule = "2 SD cutoff")

warehouse_boost_tight <- run_sector_enhancements(
  "Warehouse (1.5 SD cutoff)",
  warehouse_model_data_tight,
  warehouse_indicators
) %>%
  mutate(outlier_rule = "1.5 SD cutoff")

warehouse_boost_results <- bind_rows(
  warehouse_boost_default,
  warehouse_boost_tight
) %>%
  filter(horizon %in% candidate_horizons) %>%
  select(
    segment, outlier_rule, horizon, enhancement,
    baseline_RMSE, full_RMSE, baseline_MAPE, full_MAPE, lift_pct
)

warehouse_boost_results %>%
  mutate(
    across(ends_with("_RMSE"), ~ comma(round(.))),
    across(ends_with("_MAPE"), ~ percent(. / 100, accuracy = 0.1)),
    lift_pct = percent(lift_pct / 100, accuracy = 0.1)
) %>%
  kable(
    caption = "Warehouse Accuracy Boost Experiments"
) %>%
  kable_styling(full_width = FALSE) %>%
  scroll_box(height = "600px")
```

```{r}
# CHUNK U13C — Pick best warehouse boost

warehouse_boost_candidates <- warehouse_boost_results %>%
  filter(full_RMSE < baseline_RMSE, full_MAPE <= acceptable_mape) %>%
  arrange(full_MAPE, desc(as.integer(substr(horizon, 1, 1))))

cat("=== Warehouse candidates that beat baseline and meet MAPE ≤ 30% ===\n\n")
```

```
if (nrow(warehouse_boost_candidates) > 0) {
  warehouse_boost_candidates %>%
    mutate(
      baseline_RMSE = comma(round(baseline_RMSE)),
      full_RMSE = comma(round(full_RMSE)),
      lift_pct = percent(lift_pct / 100, accuracy = 0.1),
      full_MAPE = percent(full_MAPE / 100, accuracy = 0.1),
      baseline_MAPE = percent(baseline_MAPE / 100, accuracy = 0.1)
    ) %>%
    select(
      horizon, outlier_rule, enhancement,
      baseline_RMSE, full_RMSE, lift_pct, full_MAPE
    ) %>%
    kable(
      caption = "Warehouse Boost Candidates"
    ) %>%
    kable_styling(full_width = FALSE)
} else {
  cat("No warehouse boost version beats baseline and meets MAPE ≤ 30%.\n")
}

warehouse_reference_rmse <- 113900
warehouse_reference_lift <- 39.7
warehouse_reference_mape <- 21.2

cat("\n=== Warehouse reference from earlier summary: 6Q Full, lift 39.7%, MAPE 21.2% ===\n")

if (nrow(warehouse_boost_candidates) > 0) {
  best_candidate <- warehouse_boost_candidates %>% slice(1)

  cat("Top warehouse candidate:\n")
  cat("  Horizon      :", best_candidate$horizon, "\n")
  cat("  Outlier rule :", best_candidate$outlier_rule, "\n")
  cat("  Enhancement  :", best_candidate$enhancement, "\n")
  cat("  Full RMSE    :", comma(round(best_candidate$full_RMSE)),
      " (reference:", comma(warehouse_reference_rmse), ")\n")
  cat("  Lift         :", sprintf("%.1f%%", best_candidate$lift_pct),
      " (reference:", sprintf("%.1f%%", warehouse_reference_lift), ")\n")
  cat("  Full MAPE    :", sprintf("%.1f%%", best_candidate$full_MAPE),
      " (reference:", sprintf("%.1f%%", warehouse_reference_mape), ")\n")

  mape_difference <- warehouse_reference_mape - best_candidate$full_MAPE
  lift_difference <- best_candidate$lift_pct - warehouse_reference_lift

  cat("\n  MAPE improvement vs reference: ", sprintf("%+.1f pp", -mape_difference), "\n", sep = "")
  cat("  Lift improvement vs reference: ", sprintf("%+.1f pp", lift_difference), "\n", sep = "")

  if (mape_difference > 0 && lift_difference > 0) {
```

````
    cat("\n  VERDICT: New warehouse version is better on both MAPE and lift. Use this in the deck.\n")
  } else if (mape_difference > 0) {
    cat("\n  VERDICT: New warehouse version improves MAPE only. Consider it as a tradeoff.\n")
  } else if (lift_difference > 0) {
    cat("\n  VERDICT: New warehouse version improves lift only. Reference version may still be safer.\n")
  } else {
    cat("\n  VERDICT: Earlier warehouse version is still stronger. Keep the earlier one.\n")
  }
}
```

```{r}
# CHUNK FINAL — One Final Deck Table (clean + presentation ready)

final_deck_table <- bind_rows(
  lapply(seq_along(sector_picks), function(i) {

    segment_name <- sector_picks[[i]]$segment
    selected_horizon <- sector_picks[[i]]$horizon
    selected_variant <- sector_picks[[i]]$variant

    # If no signal
    if (is.na(selected_horizon)) {
      return(
        tibble(
          segment = segment_name,
          horizon = "No signal",
          model = "Baseline only",
          indicator = "—",
          baseline_RMSE = NA,
          enhanced_RMSE = NA,
          lift_pct = NA,
          enhanced_MAPE = NA,
          takeaway = "No reliable early signal"
        )
      )
    }

    result <- all_sector_results[[segment_name]][[selected_horizon]]
    baseline_rmse <- result$metrics$baseline$RMSE

    if (selected_variant == "Levels") {
      enhanced_rmse <- result$metrics$aug_lvl$RMSE
      enhanced_mape <- result$metrics$aug_lvl$MAPE
      indicator_used <- result$best_lvl_indicator
    } else if (selected_variant == "Change-in") {
      enhanced_rmse <- result$metrics$aug_chg$RMSE
      enhanced_mape <- result$metrics$aug_chg$MAPE
      indicator_used <- paste0("d_", result$best_chg_indicator)
````

```r
    } else {
      enhanced_rmse <- result$metrics$aug_full$RMSE
      enhanced_mape <- result$metrics$aug_full$MAPE
      indicator_used <- "All indicators"
    }

    lift <- (baseline_rmse - enhanced_rmse) / baseline_rmse * 100

    takeaway_text <- case_when(
      enhanced_mape <= 10 ~ "Strong signal",
      enhanced_mape <= 30 ~ "Usable signal",
      TRUE ~ "Weak signal"
    )

    tibble(
      segment = segment_name,
      horizon = selected_horizon,
      model = selected_variant,
      indicator = indicator_used,
      baseline_RMSE = baseline_rmse,
      enhanced_RMSE = enhanced_rmse,
      lift_pct = lift,
      enhanced_MAPE = enhanced_mape,
      takeaway = takeaway_text
    )
  })
)

final_deck_table %>%
  mutate(
    baseline_RMSE = comma(round(baseline_RMSE)),
    enhanced_RMSE = comma(round(enhanced_RMSE)),
    lift_pct = percent(lift_pct / 100, accuracy = 0.1),
    enhanced_MAPE = percent(enhanced_MAPE / 100, accuracy = 0.1)
  ) %>%
  kable(caption = "Final Segment Summary — Best Model per Segment") %>%
  kable_styling(full_width = FALSE)
```